SHERWOOD-LOGAN & ASSOCIATES, INC.

WASH. (301) 970-2181 | *2140 RENARD COURT ANNAPOLIS, MARYLAND 21401 FAX: (410) 841-6330* | *BALT: (410) 841-6810*

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

INSTRUMENTATION RESUBMITTAL

SECTION 17500 - INSTRUMENTATION

April 16, 2026

ENGINEER: WRA

CONTRACTOR: JOHNSTON CONSTRUCTION

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Table of Contents

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

General Comments:

1. This submittal assumes cable lengths for the various instruments included. Contractor to confirm cable lengths for instruments.

Resubmittal #17500-01-01 Comment Confirmations:

1. Specification Section 17500.1.03.B for non-contacting level transmitters states that calculations, arrangement, and dimensional drawings for the installation location shall be submitted. These were not included in this submittal for the Radar Level Transmitter.

   Sherwood Logan Response:

   A dimensional drawing with calculations has been included.

2. Submittal "11307-01-02 NEW Back Up Pump" includes float LSL-02 and LSH-02. This submittal also includes LSL-02 and LSH-02 on page 37/82. Contractor shall coordinate supply and provide 6 floats total, not 8.

   Sherwood Logan Response:

   The tags included in this submittal for the float switches have been updated so as not to include floats LSL-02 and LSH-02 which have already been submitted on.

3. Submit mounting hardware as indicated on Contract Drawing I-9.

   Sherwood Logan Response:

   These types of floats are “mounted” using an external weight which is supplied with the float switch.

4. After discussion with the County, the digital recorder is not needed for the site. Remove digital chart recorder cutsheet. Provide a credit associated with the removal of the digital chart recorder from the project.

   Sherwood Logan Response:

   The digital chart recorder has been removed from this submittal

5. The submitted display does not meet the requirements of Specification 17500 Section 2.05:
   A. Display shall be a 7 digit
   B. Display shall contain a bar graph

   Sherwood Logan Response:

   The display has been updated to a manufacturer and model that meets the requirements of Specification 17500 Section 2.05.

6. The specified Endress+Hauser Micropilot FMR20 is being phased out by the manufacturer. Provide the manufacturer's current replacement model FMR20B, configured to meet the specified performance and signal requirements. The replacement model is the manufacturer's current equivalent and is available at a lower cost than the originally specified unit. Revise and resubmit product data accordingly.

   Sherwood Logan Response:

   The Micropilot FMR20 that was previously submitted on has been replaced with the FMR20B in this submittal.

Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION
1	Endress + Hauser	FMR20B-FBBADTBMVCGVEED+Z1	Radar Level Transmitter
1	Endress + Hauser	71429910	Swiveling Mounting Bracket, Sewer
2	Anchor Scientific	GSE60NONC-GOLD	Float Switch, NO
4	E+H	RIA452-C211A11A	Digital Process Meter Display

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Spare Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION	QTY
1	Endress + Hauser	FMR20B-FBBADTBMVCGVEED+Z1	Radar Level Transmitter	1

32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

Data Sheet 1

<table>
<tr><td>Customer:
Reference:
Date:</td><td>Johnston Construction
32384 Charles County Pomonkey Pump Station Improvements
04/16/2026</td></tr>
<tr><td><u>Qty</u></td><td><u>Description</u></td></tr>
<tr><td>1</td><td>Mfg: Endress + Hauser: Radar Level Transmitter
Model Number: FMR20B-FBBADTBMVCGVEED+Z1

Tags/Service:
LT-01 / Pump Station Wet Well

Specifications:
Up to 66ft measuring range
FMR20 - Micropilot FMR20 series time-of-flight radar measurement
FB - CAN/US IS Cl.I,II,III Div.1 Gr.A-G T4 AEx/Ex ia IIC T4, AEx/Ex ia IIIC
BA - 2 wire, 4-20mA HART output
D - LED + bluetooth display/operation
T - Pre-installed cable for electrical connection
BM - PVDF encapsulated, 40mm / 1-½“ antenna
VCG - Thread ASME MNPT1 process connection cable entry
VEE - Thread ASME MNPT1-½ process connection antenna end, PVDF material
D - 20m/65ft cable length
Z1 - Tagging</td></tr>
<tr><td>1</td><td>Mfg: Endress + Hauser: Swiveling Mounting Bracket, Sewer
Model Number: 71429910

Tags/Service:</td></tr>
</table>

	LT-01 / Pump Station Wet Well Specifications: • Pivotable and adjustable arm for mounting E+H FMR20B • Allows alignment of transmitter with the center of a channel

TI01796F/00/EN/04.25-00
71726356
2025-09-30

RADAR LEVEL TRANSMITTER

TAGS: LT-01

Technical Information
Micropilot FMR20B

Free-space radar
HART

Level measurement in liquids and bulk solids

Application

- Continuous, non-contact level measurement of liquids and bulk solids
- Degree of protection: IP66/68 / NEMA Type 4X/6P
- Maximum measuring range up to 30 m (98 ft)
- Process temperature: −40 to 80 °C (−40 to 176 °F)
- Process pressure: −1 to 3 bar (−14 to 43 psi)
- Accuracy: Up to ± 2 mm (0.08 in)
- International explosion protection certificates

Your benefits

- LED indicator for fast status detection
- Easy, guided commissioning with intuitive user interface
- Radar measuring device with *Bluetooth*® wireless technology and HART communication
- Easy, reliable and encrypted wireless remote access – ideal for difficult-to-reach installations, even in hazardous areas
- Commissioning, operation and maintenance via free iOS/Android app SmartBlue – saves time and reduces costs
- Flow measurement in open channels or weirs with totalizer

Endress+Hauser
People for Process Automation

Table of contents

About this document

Symbols

Safety symbols

⚠ DANGER
This symbol alerts you to a dangerous situation. Failure to avoid this situation will result in serious or fatal injury.

⚠ WARNING
This symbol alerts you to a potentially dangerous situation. Failure to avoid this situation can result in serious or fatal injury.

⚠ CAUTION
This symbol alerts you to a potentially dangerous situation. Failure to avoid this situation can result in minor or medium injury.

NOTICE
This symbol alerts you to a potentially harmful situation. Failure to avoid this situation can result in damage to the product or something in its vicinity.

Communication-specific symbols

Bluetooth®:
Wireless data transmission between devices over a short distance via radio technology.

Symbols for certain types of information

Permitted:

Procedures, processes or actions that are permitted.

Forbidden:

Procedures, processes or actions that are forbidden.

Additional information:

Reference to documentation:

Reference to page:

Series of steps: 1., 2., 3.

Result of an individual step: ↳

Symbols in graphics

Item numbers: 1, 2, 3 ...

Series of steps: 1., 2., 3.

Views: A, B, C, ...

List of abbreviations

PN
Nominal pressure

MWP
Maximum working pressure
The maximum working pressure is indicated on the nameplate.

ToF
Time of Flight

DTM
Device Type Manager

ε_r (Dk value)
Relative dielectric constant

Operating tool

The term "operating tool" is used in place of the following operating software:
- FieldCare / DeviceCare, for operation via HART communication, IO-Link communication and PC
- SmartBlue app for operation using an Android or iOS smartphone or tablet

PLC
Programmable logic controller (PLC)

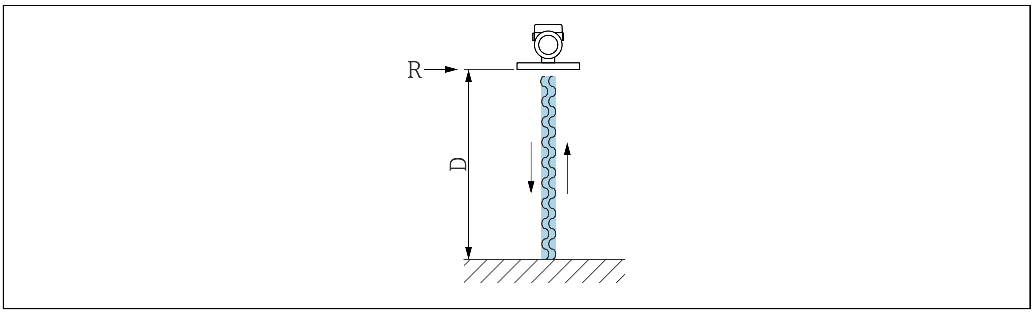
Graphic conventions

- Installation, explosion and electrical connection drawings are presented in simplified format
- Devices, assemblies, components and dimensional drawings are presented in reduced-line format
- Dimensional drawings are not to-scale representations; the dimensions indicated are rounded off to 2 decimal places
- Unless otherwise described, flanges are presented with sealing surface form EN 1092-1; ASME B16.5, RF.

Function and system design

Measuring principle

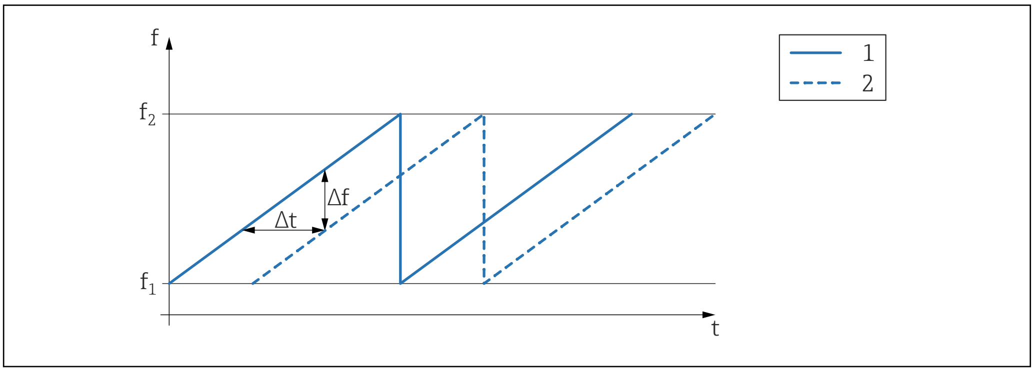
The Micropilot is a "downward-looking" measuring system, operating based on the frequency modulated continuous wave method (FMCW). The antenna emits an electromagnetic wave at a continuously varying frequency. This wave is reflected by the product and received again by the antenna.



1 FMCW principle: Transmission and reflection of the continuous wave

R Reference point of measurement
D Distance between reference point and product surface

The frequency of this wave is modulated in the form of a sawtooth signal between two limit frequencies f_1 and f_2:



2 FMCW principle: Result of frequency modulation

1 Transmitted signal
2 Received signal

This results in the following difference frequency at any time between the transmitted signal and the received signal:

$\Delta f = k \, \Delta t$

where Δt is the run time and k is the specified increase in frequency modulation.

Δt is given by the distance D between the reference point R and the product surface:

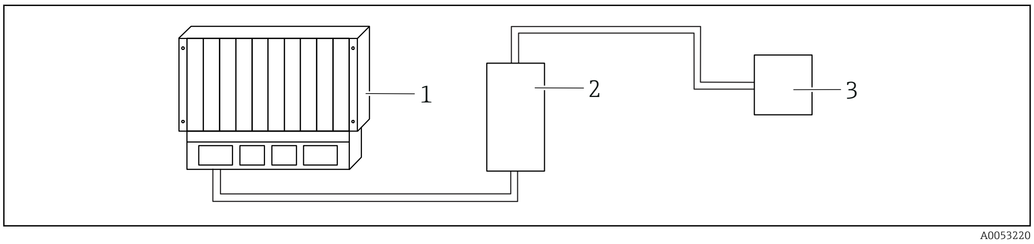
$D = (c \, \Delta t) / 2$

where c is the wave velocity.

In summary, D can be calculated from the measured difference frequency Δf. D is then used to determine the fill level or flow rate.

Measuring system

A complete measuring system comprises:



1 PLC (programmable logic controller)
2 RMA42/RIA45 (if necessary)
3 Device

Communication and data processing

- 4 to 20 mA with superimposed digital communication protocol HART, 2-wire
- Bluetooth® wireless technology (optional)

Dependability

IT security

The manufacturer warranty is valid only if the product is installed and used as described in the Operating Instructions. The product is equipped with security mechanisms to protect it against any inadvertent changes to the settings.

IT security measures, which provide additional protection for the product and associated data transfer, must be implemented by the operators themselves in line with their security standards.

Device-specific IT security

The device offers specific functions to support protective measures by the operator. These functions can be configured by the user and guarantee greater in-operation safety if used correctly. The user role can be changed with an access code (applies to operation via Bluetooth® wireless technology or FieldCare, DeviceCare, Asset Management Tools (e.g. AMS, PDM)).

Access via Bluetooth® wireless technology

Secure signal transmission via Bluetooth® wireless technology uses an encryption method tested by the Fraunhofer Institute.

- Without the SmartBlue app, the device is not visible via Bluetooth® wireless technology.
- Only one point-to-point connection is established between the device and a smartphone or tablet.
- The Bluetooth® interface can be disabled via SmartBlue or an operating tool via digital communication.

Input

Measured variable

The measured variable is the distance between the reference point and the product surface.

The level is calculated based on **E**, the empty distance entered.

Measuring range

The measuring range starts at the point where the beam hits the tank floor. Levels below this point cannot be measured, particularly in the case of spherical bases or conical outlets.

Maximum measuring range

The maximum measuring range depends on the antenna size.

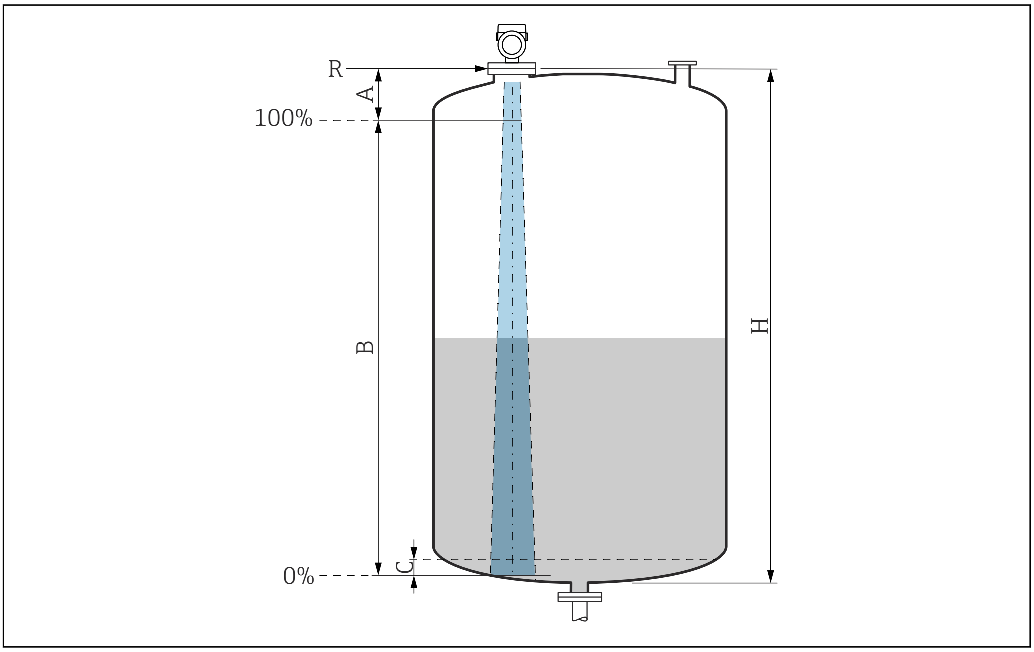
Antenna	Maximum measuring range
40 mm (1.5 in)	20 m (65.6 ft)
~~80 mm (3 in)~~	~~30 m (98.4 ft)~~

Usable measuring range

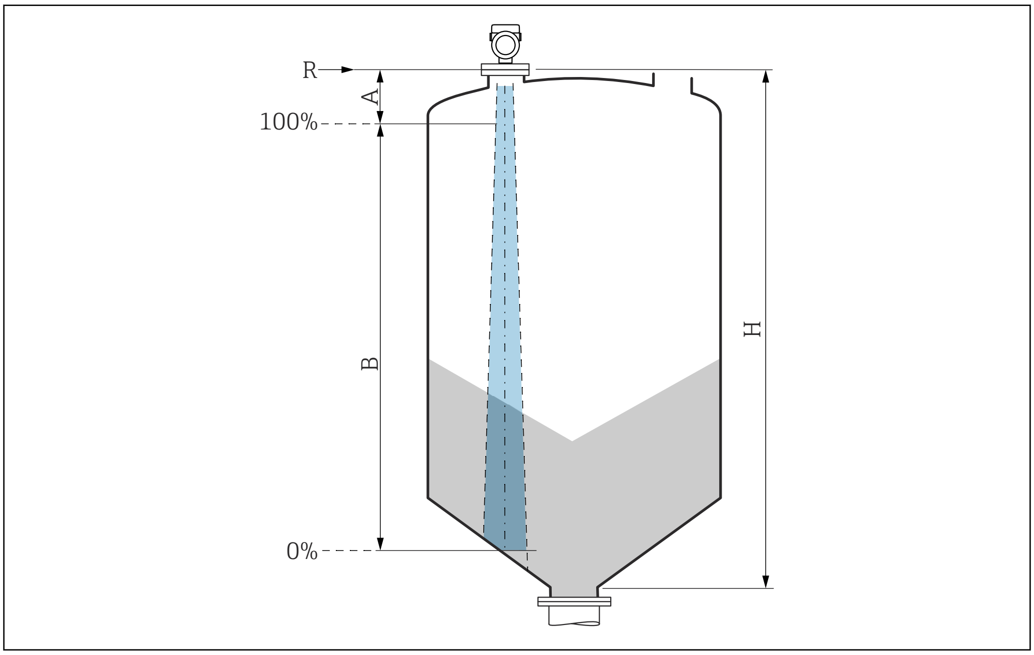
The usable measuring range depends on the antenna size, the medium's reflective properties, the installation position and any possible interference reflections.

In principle, measurement is possible up to the tip of the antenna.

Depending on the position of the product (angle of repose for solids) and to avoid any material damage from corrosive or aggressive media or deposit buildup on the antenna, the end of the measuring range should be selected 10 mm (0.4 in) before the tip of the antenna.



A Antenna tip + 10 mm (0.4 in)
B Usable measuring range
C 50 to 80 mm (1.97 to 3.15 in); Medium ε_r ≤2
H Vessel height
R Reference point of the measurement, varies depending on the antenna system (see section on Mechanical construction)



A0051659

A *Antenna tip + 10 mm (0.4 in)*
B *Usable measuring range*
H *Vessel height*
R *Reference point of the measurement, varies depending on the antenna system (see section on Mechanical construction)*

In the case of media with a low dielectric constant ε_r <2, the tank bottom may be visible through the medium when levels are very low (less than level C). Reduced accuracy must be expected in this range. If this is not acceptable, the zero point should be positioned at a distance C above the tank bottom for these applications (see figure).

The media groups and the possible measuring range are described as a function of the application and media group in the following section. If the relative permittivity of the medium is not known, to ensure a reliable measurement assume the medium belongs to group B.

Media groups

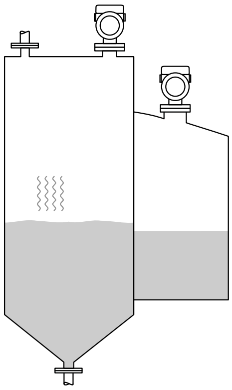
- **A** (ε_r 1.4 to 1.9)
  Non-conductive liquids, e.g. liquefied gas
- **B** (ε_r 1.9 to 4)
  Non-conductive liquids, e.g. gasoline, oil, toluene, etc.
- **C** (ε_r 4 to 10)
  e.g. concentrated acid, organic solvents, ester, aniline, etc.
- **D** (ε_r >10)
  Conductive liquids, aqueous solutions, diluted acids, bases and alcohol

For the relative permittivity values (ε_r values) of many media commonly used in industry, please refer to:
- Relative permittivity (ε_r value), Compendium CP01076F
- The Endress+Hauser "DC Values app" (available for Android and iOS)

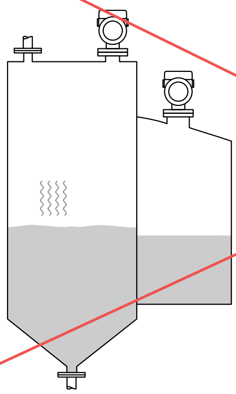
Measurement in storage vessel

Storage vessel - measuring conditions
Calm medium surface (e.g. bottom filling, filling via immersion tube or rare filling from above)

40 mm (1.5 in) antenna in storage vessel

	Media group	Measuring range
	**A** (ε_r 1.4 to 1.9)	10 m (33 ft)
	B (ε_r 1.9 to 4)	20 m (65.6 ft)
	C (ε_r 4 to 10)	20 m (65.6 ft)
	D (ε_r >10)	20 m (65.6 ft)

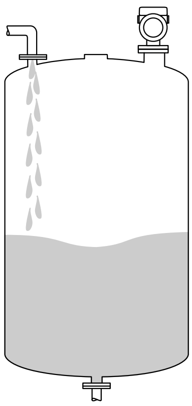
80 mm (3 in) antenna in storage vessel

	Media group	Measuring range
	**A** (ε_r 1.4 to 1.9)	12 m (39 ft)
	B (ε_r 1.9 to 4)	23 m (75 ft)
	C (ε_r 4 to 10)	30 m (98 ft)
	D (ε_r >10)	30 m (98 ft)

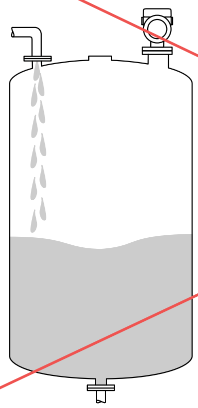
Measurement in buffer vessel

Buffer vessel - measuring conditions
Moving medium surface (e.g. permanent free filling from above, mixing jets)

40 mm (1.5 in) antenna in buffer vessel

	Media group	Measuring range
	**A** (ε_r 1.4 to 1.9)	7 m (23 ft)
	B (ε_r 1.9 to 4)	13 m (43 ft)
	C (ε_r 4 to 10)	20 m (65.6 ft)
	D (ε_r >10)	20 m (65.6 ft)

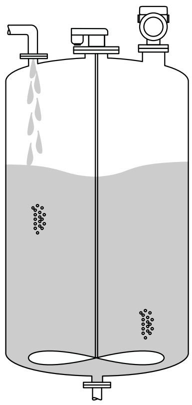
80 mm (3 in) antenna in buffer vessel

	Media group	Measuring range
	**A** (ε_r 1.4 to 1.9)	7.5 m (25 ft)
	B (ε_r 1.9 to 4)	15 m (49 ft)
	C (ε_r 4 to 10)	28 m (92 ft)
	D (ε_r >10)	30 m (98 ft)

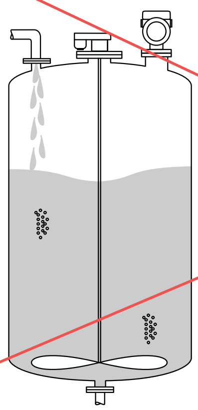
Measurement in vessel with agitator

Vessel with agitator - measuring conditions
Turbulent medium surface (e.g. from filling from above, stirrers and baffles)

40 mm (1.5 in) antenna in vessel with agitator

	Media group	Measuring range
	**A** (ε_r 1.4 to 1.9)	4 m (13 ft)
	B (ε_r 1.9 to 4)	5 m (16.4 ft)
	C (ε_r 4 to 10)	13 m (43 ft)
	D (ε_r >10)	20 m (65.6 ft)

80 mm (3 in) antenna in vessel with agitator

	Media group	Measuring range
	**A** (ε_r 1.4 to 1.9)	4 m (13 ft)
	B (ε_r 1.9 to 4)	7 m (23 ft)
	C (ε_r 4 to 10)	15 m (49 ft)
	D (ε_r >10)	25 m (82 ft)

Operating frequency

approx. 80 GHz

Up to eight devices can be installed in a tank without the devices mutually influencing one another.

Transmission power

- Peak power: <1.5 mW
- Average output power: <70 µW

Output

Output signal

- 4 to 20 mA with superimposed digital communication protocol HART, 2-wire
- The current output offers a choice of three different operating modes:
  - 4 to 20.5 mA
  - NAMUR NE 43: 3.8 to 20.5 mA (factory setting)
  - US mode: 3.9 to 20.5 mA

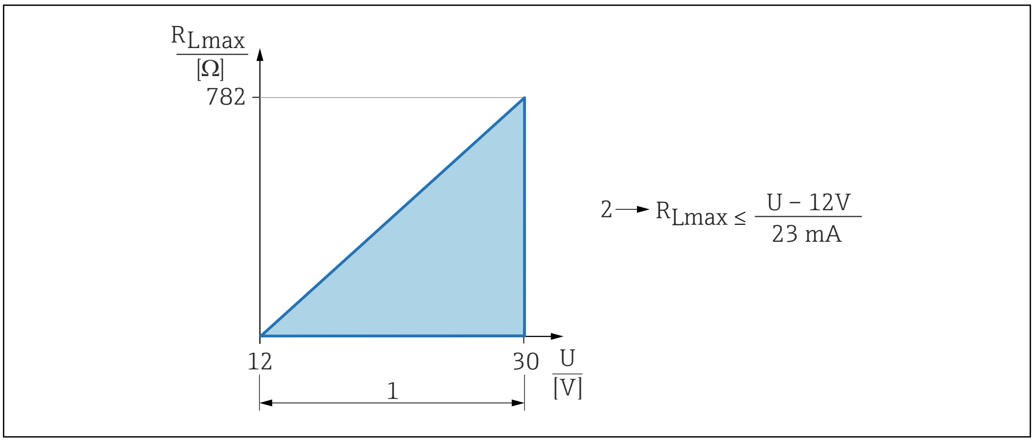
Signal on alarm for devices with current output

Current output
Signal on alarm in accordance with NAMUR recommendation NE 43.
- Max. alarm: can be set from 21.5 to 23 mA
- Min. alarm: < 3.6 mA (factory setting)

Operating tool via digital communication
Status signal (according to NAMUR Recommendation NE 107):
Plain text display

Load

In order to guarantee sufficient terminal voltage, a maximum load resistance R L (including line resistance) must not be exceeded, depending on the supply voltage U of the supply unit.



A0052602

1 Power supply 12 to 30 V
2 R_{Lmax} maximum load resistance
U Supply voltage

If load is too great:
- Failure current is indicated and error message is displayed (indication: MIN alarm current)
- Periodic checking to establish if it is possible to quit fault state

Operation via handheld terminal or PC with operating program: take minimum communication resistor of 250 Ω into consideration.

Damping

Damping affects all continuous outputs.
Factory setting: 0 s(can be set from 0 to 999 s)

Ex connection data

See the separate technical documentation (Safety Instructions (XA)) on www.endress.com/download.

Linearization

The linearization function of the device allows the conversion of the measured value into any unit of length, weight, flow or volume.

Pre-programmed linearization curves

Linearization tables for calculating the volume in the following vessels are pre-programmed into the device:
- Pyramid bottom
- Conical bottom
- Angled bottom
- Horizontal cylinder
- Sphere

Linearization tables for calculating the flow rate are pre-programmed into the device and include the following:
- Flumes
  - Khafagi Venturi flume
  - Venturi flume
  - Parshall flume
  - Palmer Bowlus flume
  - Trapezoidal flume (ISO 4359)
  - Rectangular flume (ISO 4359)
  - U-shaped flume (ISO 4359)
- Weirs
  - Trapezoidal weir
  - Rectangular broad-crested weir (ISO 3846)
  - Rectangular thin-plate weir (ISO 1438)
  - V-notch thin-plate weir (ISO 1438)
- Standard formula

Other linearization tables of up to 32 value pairs can be entered manually.

For further information on flow measurement over open channels and weirs, see SD03445F.

Totalizer

The device offers a totalizer which adds up the flow rate. The totalizer cannot be reset.

Protocol-specific data

Manufacturer ID:
17(0x0011)

Device type ID:
0x11DE

Device revision:
2

HART specification:
7.6

DD version:
1

Device description files (DTM, DD)
Information and files at:
- www.endress.com
  On the product page for the device: Documents/Software → Device drivers
- www.fieldcommgroup.org

HART load:
Min. 250 Ω

The following measured values are assigned to the device variables at the factory:

Device variable	Measured value
Primary variable (PV) [1]	Level linearized
Secondary variable (SV)	Distance
Tertiary variable (TV)	Absolute echo amplitude
Quaternary variable (QV)	Relative echo amplitude

1) The PV is always applied to the current output.

Choice of HART device variables
- Level linearized
- Distance
- Electronics temperature
- Sensor temperature
- Absolute echo amplitude
- Relative echo amplitude
- Area of incoupling
- Percent of range
- Loop current
- Flow
- Totalizer value
- Not used

Supported functions

- Burst mode
- Additional transmitter status
- Device locking

Wireless HART data

Minimum start-up voltage:
12 V

Start-up current:
< 3.6 mA

Starting time:
< 15 s

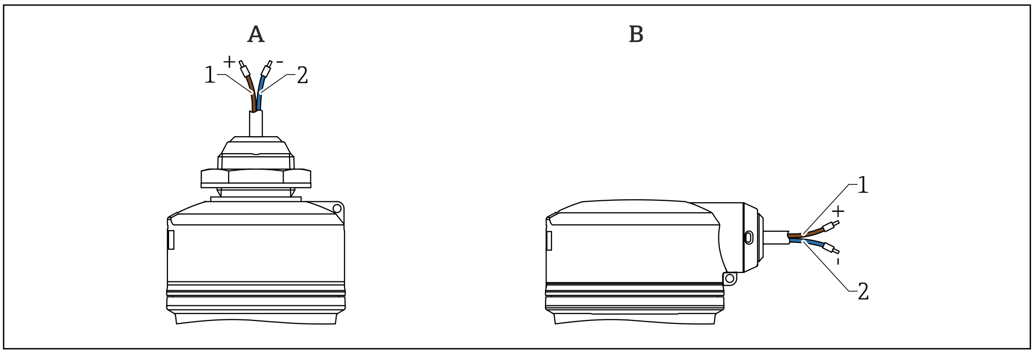
Minimum operating voltage:
12 V

Multidrop current:
4 mA

Time to establish connection:
< 30 s

Power supply

Cable assignment



3 *Cable assignment*

A Cable entry from above
B Side cable entry
1 Plus, brown wire
2 Minus, blue wire

Supply voltage

12 to 30 V DC on a DC power unit

The power unit must be safety-approved (e.g. PELV, SELV, Class 2) and must comply with the relevant protocol specifications.

Protective circuits against reverse polarity, HF influences and overvoltage peaks are installed.

Power consumption

- Non-hazardous area: To meet device safety specifications according to the IEC 61010 standard, the installation must ensure that the maximum current is limited to 500 mA.
- Hazardous area: The maximum current is restricted to Ii = 100 mA by the transmitter power supply unit when the measuring instrument is used in an intrinsically safe circuit (Ex ia).

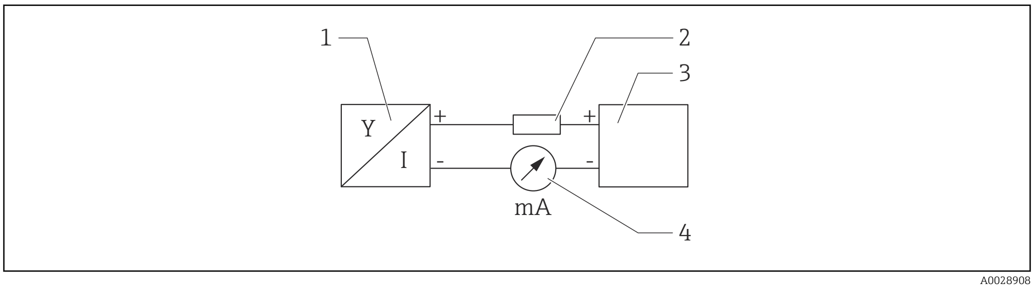
Potential equalization

No special measures for potential equalization are required.

Connecting the device

Function diagram of 4 to 20 mA HART

Connection of the device with HART communication, power source and 4 to 20 mA indicator



4 Function diagram of HART connection

1 *Device with HART communication*
2 *HART resistor*
3 *Power supply*
4 *Multimeter or ammeter*

The HART communication resistor of 250 Ω in the signal line is always necessary in the case of a low-impedance power supply.

The voltage drop to be taken into account is:
Max. 6 V for 250 Ω communication resistor

Function diagram of HART device, connection with RIA15, display only without operation, without communication resistor

The RIA15 remote display can be ordered together with the device.

Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

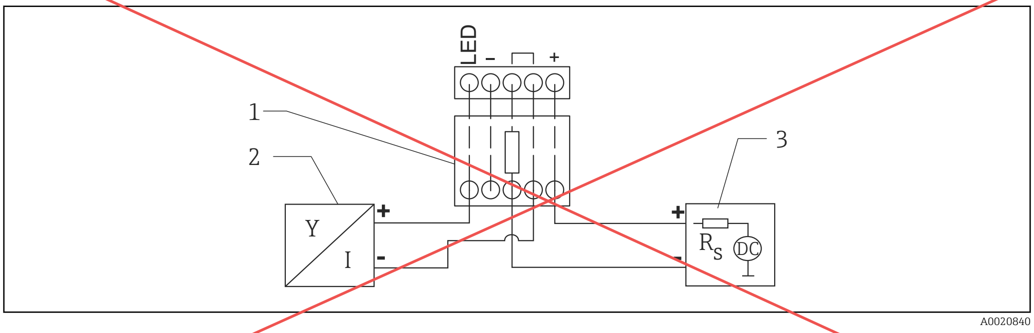
Terminal assignment RIA15

- **+**
  Positive connection, current measurement
- **-**
  Negative connection, current measurement (without backlighting)
- **LED**
  Negative connection, current measurement (with backlighting)
- ⏚
  Functional grounding: Terminal in housing

The RIA15 process indicator is loop-powered and does not require any external power supply.

The voltage drop to be taken into account is:
- ≤1 V in the standard version with 4 to 20 mA communication
- ≤1.9 V with HART communication
- and an additional 2.9 V if display light is used

Connection of the HART communication resistor module, RIA15 with backlighting



8 *Function diagram of HART device, RIA15 with light, HART communication resistor module*

1 *HART communication resistor module*
2 *Device with HART communication*
3 *Current supply*

Cable specification

Unshielded cable, wire cross-section 0.5 mm²
- Resistant to UV and weather conditions as per ISO 4892-2
- Flame resistance according to IEC 60332-1-2

As per IEC 60079-11 section 9.4.4, the cable is designed for a tensile strength of 30 N (6.74 lbf) (over a period of 1 h).

The device is available in cable lengths of 5 m (16 ft), 10 m (32 ft), 15 m (49 ft), 20 m (65 ft), 30 m (98 ft) and 50 m (164 ft).

User-defined lengths up to total length of 300 m (980 ft) are possible in increments of one meter (order option "1") or one foot (order "2").

For devices with marine approval:
- Only available with a length of 10 m (32 ft) and "user-defined"
- Halogen-free as per IEC 60754-1
- No development of corrosive fire gases in accordance with IEC 60754-2
- Low flue gas density in accordance with IEC 61034-2

Overvoltage protection

The device satisfies the IEC/DIN EN 61326-1 product standard (Table 2 Industrial environment). Depending on the type of connection (DC power supply, input line, output line), different test levels are used to prevent transient overvoltages (IEC/DIN EN 61000-4-5 Surge) in accordance with IEC/DIN EN 61326-1: Test level for DC power supply lines and IO lines: 1 000 V wire to ground.

Devices for the "protection by enclosure" explosion protection are equipped with an integrated overvoltage protection system.

Overvoltage category

In accordance with IEC/DIN EN 61010-1, the device is intended for use in networks with overvoltage protection category II.

Performance characteristics

Reference operating conditions

- As per IEC 62828-1/IEC 62828-4
- Ambient temperature T_A = constant, in the range of +21 to +33 °C (+70 to +91 °F)
- Humidity φ = constant, in the range of: 5 to 80 % RH ± 5 %
- Atmospheric pressure p_U = constant, in the range of: 860 to 1 060 mbar (12.47 to 15.37 psi)
- Load with HART: 250 Ω
- Supply voltage: DC 24 V ±3 V
- Reflector: metal plate with diameter ≥ 1 m (40 in)
- No major interference echoes inside the signal beam

Response time

- HART: acyclic: min. 330 ms, typically 590 ms (depending on commands and number of preambles)
- HART: cyclic (burst): min. 160 ms, typically 350 ms (depending on commands and number of preambles)

Resolution

Current output: < 1 µA

Digital: 1 mm (0.04 in)

Maximum measurement error

Reference accuracy

Accuracy

The accuracy is the sum of the non-linearity, non-repeatability and hysteresis.

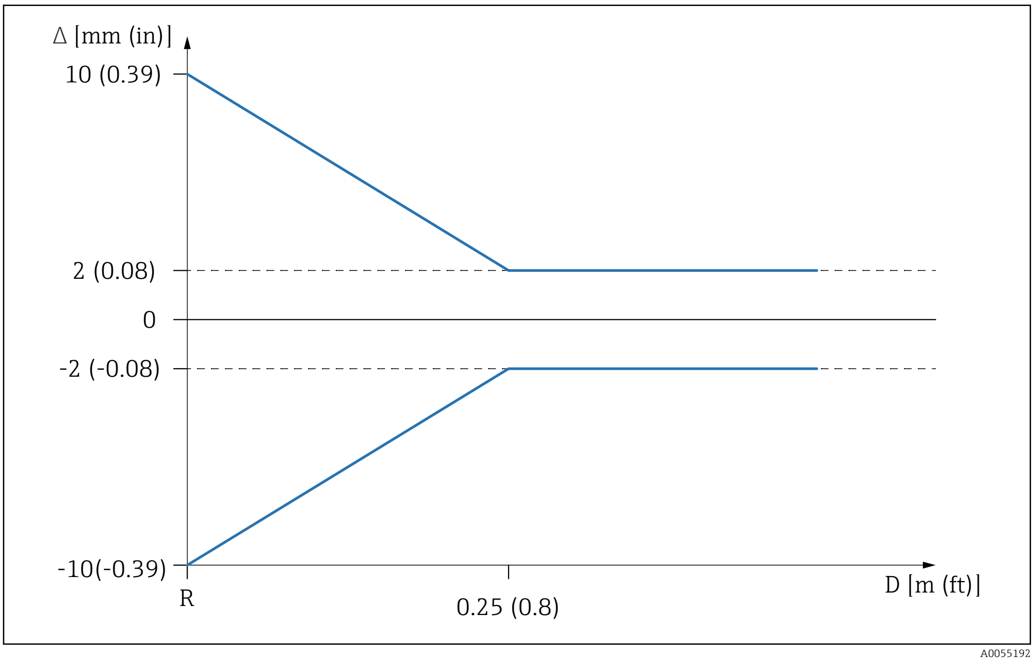
For liquids:
- Measuring distance up to 0.25 m (0.82 ft): max. ±10 mm (±0.39 in)
- Measuring distance > 0.25 m (0.82 ft): ±2 mm (±0.08 in)

For solids:
- Measuring distance up to 0.8 m (2.6 ft): max. ±20 mm (±0.79 in)
- Measuring distance > 0.8 m (2.6 ft): ±4 mm (±0.16 in)

Non-repeatability
Non-repeatability is already included in the accuracy.
≤ 1 mm (0.04 in)

If conditions deviate from the reference operating conditions, the offset/zero point that results from the installation conditions can be up to ±4 mm (±0.16 in). This additional offset/zero point can be eliminated by entering a correction (**Level correction** parameter) during commissioning.

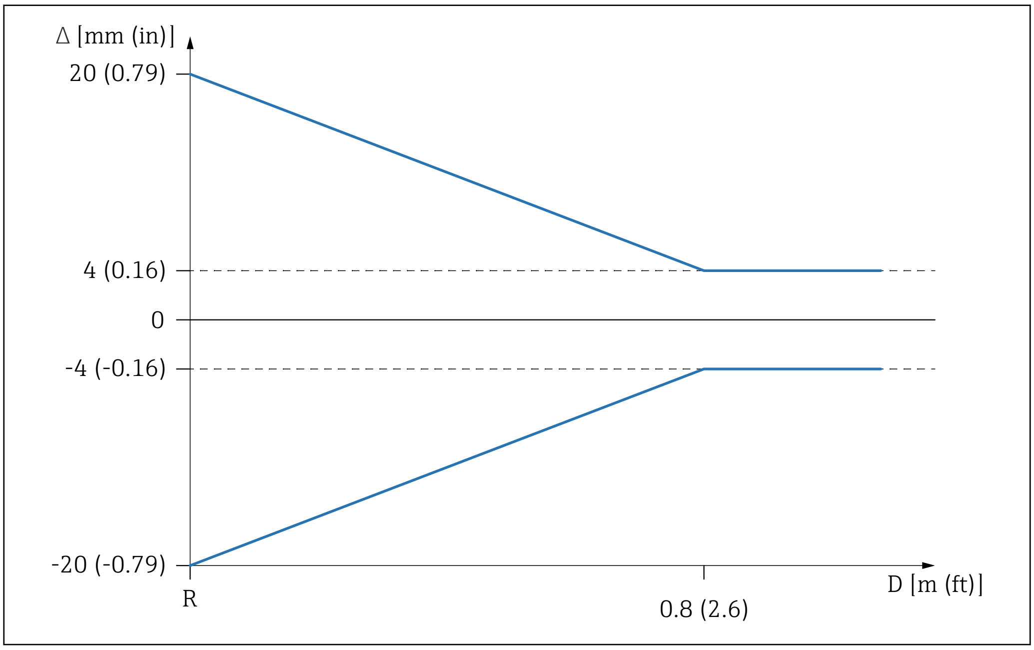
Differing values in near-range applications for liquids



9 Maximum measurement error in near-range applications

Δ Maximum measurement error
R Reference point of the distance measurement
D Distance from reference point of antenna

Differing values in near-range applications for solids



10 Maximum measurement error in near-range applications

Δ Maximum measurement error
R Reference point of the distance measurement
D Distance from reference point of antenna

Influence of ambient temperature

The output changes due to the effect of the ambient temperature with respect to the reference temperature.

The measurements are performed according to IEC 61298-3 / IEC 60770-1

Digital output (HART)
Average T_C = ±2 mm (±0.08 in)/10 K

Analog (current output)
- Zero point (4 mA): average T_C = 0.02 %/10 K
- Span (20 mA): average T_C = 0.05 %/10 K

Reaction time

According to IEC 61298-2 / IEC 60770-1 , the step response time is the time following an abrupt change in the input signal up until the changed output signal has adopted 90 % of the steady-state value for the first time.

The response time can be configured.

The following step response times apply (in accordance with IEC 61298-2/IEC 60770-1) when damping is switched off:
- Measuring rate ≤ 250 ms at operating voltage 24 V
- Step response time < 1 s

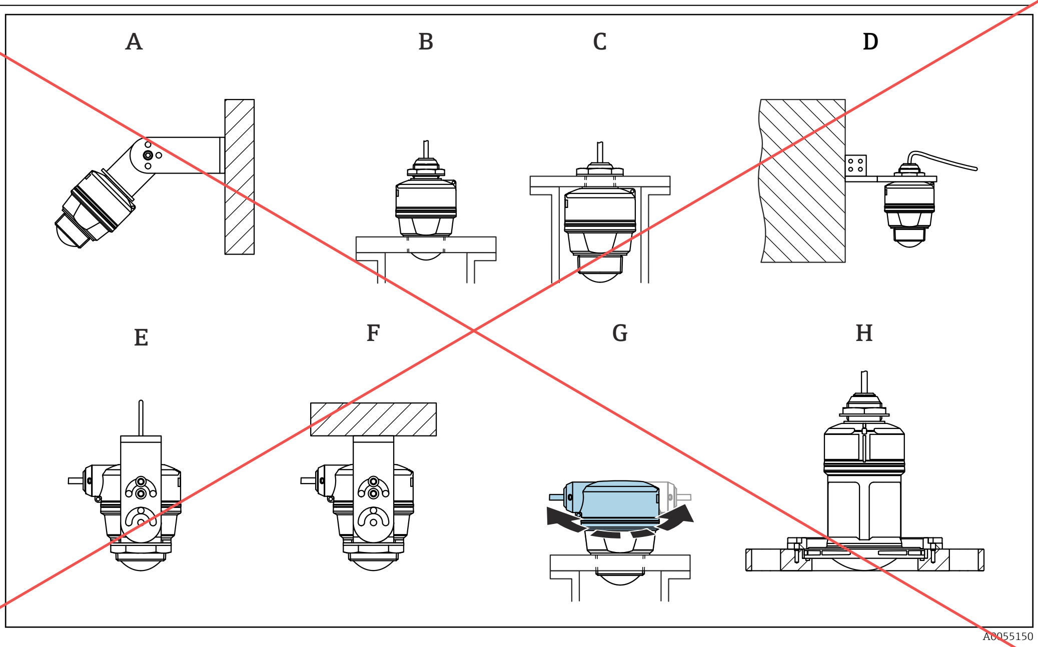
Warm-up time

The warm-up time (in accordance with IEC 62828-4) indicates the time required for the device to reach its maximum accuracy or performance after the supply voltage is energized.

Warm-up time: ≤ 15 s

Installation

Installation types

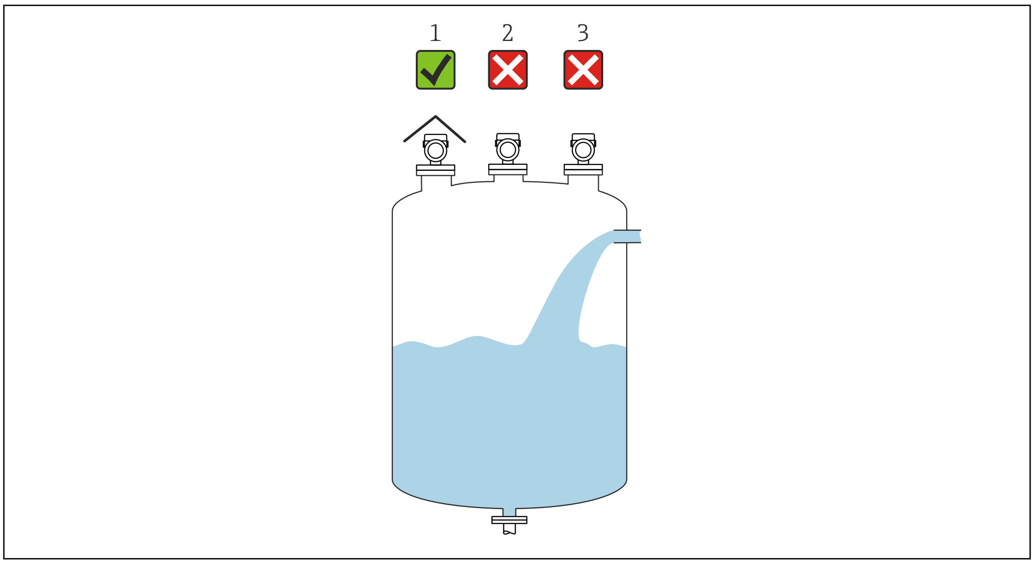


11 *Wall or ceiling mount*

A Wall mounting adjustable
B Tightened at antenna end process connection
C Tightened at cable entry from above process connection
D Wall mounting with cable entry from above process connection
E Rope mounting with cable entry at the side
F Ceiling mounting with cable entry at the side
G Cable entry at the side, top housing section can be rotated
H Mounting with UNI slip-on flange

Please note the following:
- The sensor cables are not designed as supporting cables. Do not use them for suspension purposes.
- For rope mounting, the rope must be provided by the customer.
- Always operate the device in a vertical position in free-space applications.
- For devices with side cable outlet and 80 mm antenna, installation is only possible with a UNI slip-on flange.

Mounting location

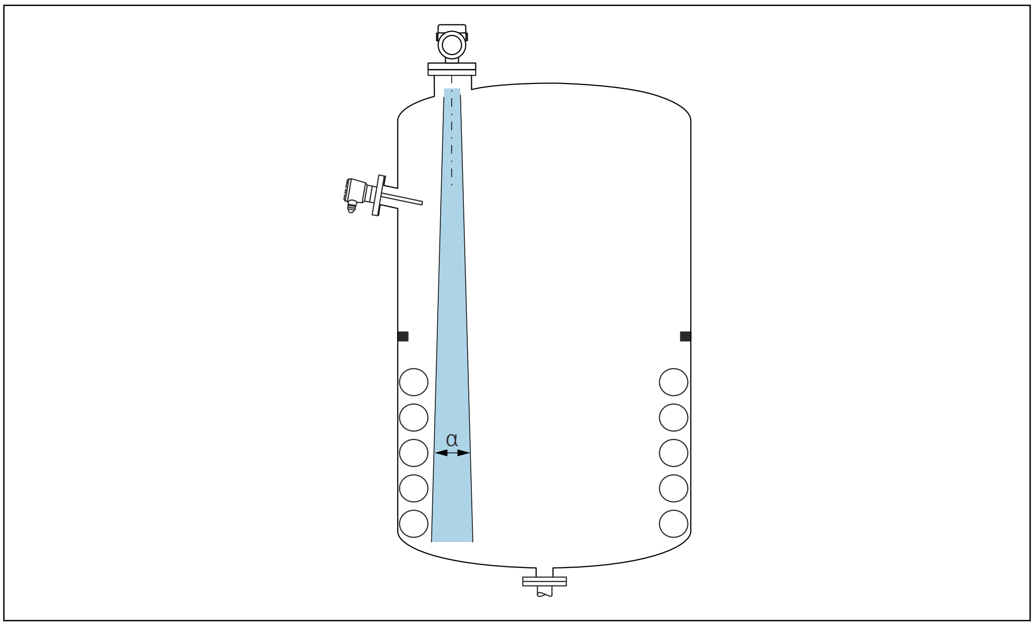


A0055811

1 *Use of a weather protection cover; protection from direct sunlight or rain*
2 *Installation not centered: Interferences can lead to incorrect signal analysis*
3 *Do not install above the filling curtain*

Orientation

Internal vessel fittings



A0031777

Avoid internal fittings (level switches, temperature sensors, struts, vacuum rings, heating coils, baffles etc.) inside the signal beam. Pay attention to the beam angle **α**.

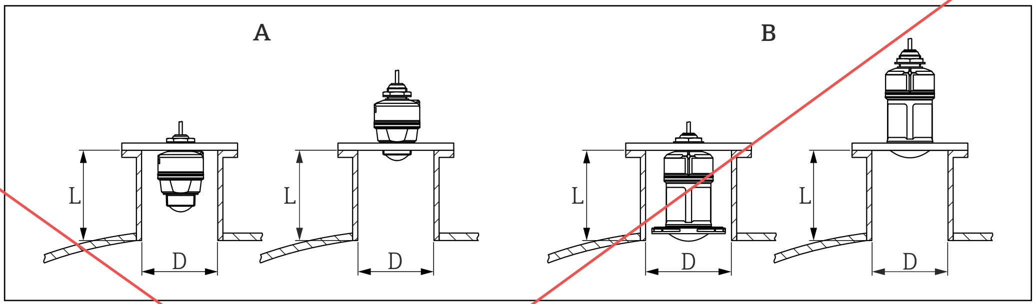
Vertical alignment of antenna axis

Align the antenna so that it is perpendicular to the product surface.

The maximum reach of the antenna can be reduced, or additional interference signals can occur, if the antenna is not installed perpendicular to the product.

Installation instructions

To ensure optimum measurement, the antenna must protrude from the nozzle. The interior of the nozzle must be smooth and must not contain any edges or welded joints. If possible, round the nozzle edge.



12 *Nozzle installation*

A 40 mm (1.5 in) antenna
B 80 mm (3 in) antenna

The maximum nozzle length **L** depends on the nozzle diameter **D**.

Please note the limits for the diameter and length of the nozzle.

40 mm (1.5 in) antenna, installation outside nozzle
- D: min. 40 mm (1.5 in)
- L: max. (D - 30 mm (1.2 in)) × 7.5

40 mm (1.5 in) antenna, installation inside nozzle
- D: min. 80 mm (3 in)
- L: max. 100 mm (3.94 in) + (D - 30 mm (1.2 in)) × 7.5

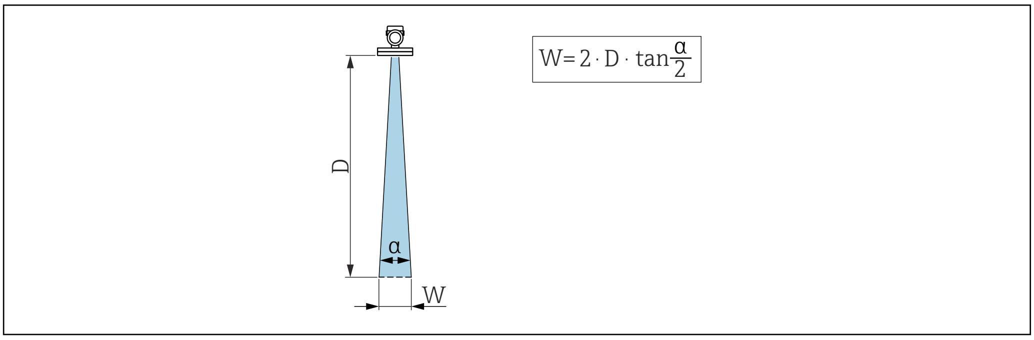
80 mm (3 in) antenna, installation inside nozzle
- D: min. 120 mm (4.72 in)
- L: max. 140 mm (5.51 in) + (D - 50 mm (2 in)) × 12

80 mm (3 in) antenna, installation outside nozzle
- D: min. 80 mm (3 in)
- L: max. (D - 50 mm (2 in)) × 12

Beam angle

Calculation

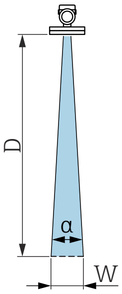
The beam angle is defined as the angle α at which the energy density of the radar waves reaches half the value of the maximum energy density (3dB width). Microwaves are also emitted outside the signal beam and can be reflected off interfering installations.



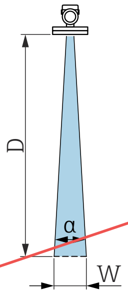
13 *Relationship between beam angle α, distance D and beamwidth diameter W*

The beamwidth diameter **W** depends on the beam angle **α** and the distance **D**.

Antenna 40 mm (1.5 in), α = 8 °

W = D × 0.14	D	W
	5 m (16 ft)	0.70 m (2.29 ft)
	10 m (33 ft)	1.40 m (4.58 ft)
	15 m (49 ft)	2.09 m (6.87 ft)
	20 m (66 ft)	2.79 m (9.16 ft)

Antenna 80 mm (3 in), α = 4 °

W = D × 0.07	D	W
	5 m (16 ft)	0.35 m (1.15 ft)
	10 m (33 ft)	0.70 m (2.30 ft)
	15 m (49 ft)	1.05 m (3.45 ft)
	20 m (66 ft)	1.40 m (4.59 ft)
	25 m (82 ft)	1.75 m (5.74 ft)
	30 m (98 ft)	2.10 m (6.89 ft)

Special installation instructions

External measurement through plastic cover or dielectric windows

- Dielectric constant of medium: $\varepsilon_r \geq 10$
- The distance from the tip of the antenna to the tank should be approx. 100 mm (4 in).
- Avoid installation positions where condensate or buildup can form between the antenna and the vessel
- In the case of outdoor installations, ensure that the area between the antenna and the tank is protected from the weather
- Do not install any fittings or attachments between the antenna and the tank that could reflect the signal

The thickness of the tank ceiling or the dielectric window depends on the ε_r of the material.

The material thickness can be a full multiple of the optimum thickness (table); it is important to note, however, that the microwave transparency decreases significantly with increasing material thickness.

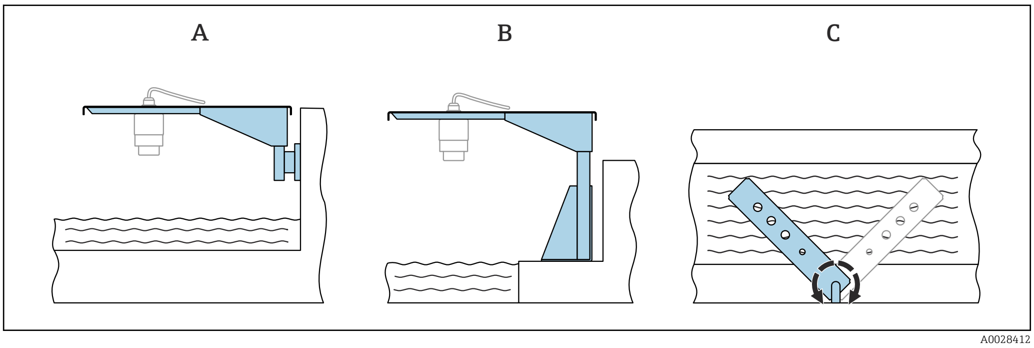
Optimum material thickness

Material	Optimum material thickness
PE; ε_r 2.3	1.25 mm (0.049 in)
PTFE; ε_r 2.1	1.30 mm (0.051 in)
PP; ε_r 2.3	1.25 mm (0.049 in)
Perspex; ε_r 3.1	1.10 mm (0.043 in)

Weather protection cover

A weather protective cover is recommended for outdoor use.

The weather protective cover can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



18 *Cantilever installation, with pivot*

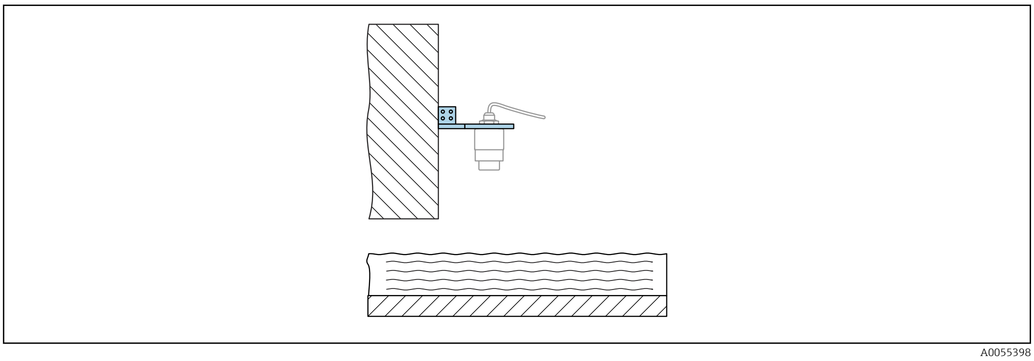
A Cantilever with wall bracket (side view)
B Cantilever with mounting frame (side view)
C Cantilever can be turned, e.g. in order to position the device over the center of the flume (top view)

NOTICE

There is no conductive connection between the mounting bracket and transmitter housing.
Electrostatic charging possible.
▶ Integrate the mounting bracket in the local potential equalization system.

Mounting with a pivotable mounting bracket

The pivotable mounting bracket can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



19 *Pivotable and adjustable cantilever with wall bracket (e.g. to align the device with the center of a flume)*

NOTICE

There is no conductive connection between the mounting bracket and transmitter housing.
Electrostatic charging possible.
▶ Integrate the mounting bracket in the local potential equalization system.

Environment

Ambient temperature range

−40 to +80 °C (−40 to +176 °F)

If operating outdoors in strong sunlight:
- Mount the device in the shade.
- Avoid direct sunlight, particularly in warmer climatic regions.
- Use a weather protective cover.

Storage temperature

−40 to +80 °C (−40 to +176 °F)

Climate class

According to IEC 60068-2-38 test Z/AD (relative humidity 4 to 100 %).

Operating height

Up to 5 000 m (16 404 ft) above sea level

Degree of protection	Testing according to IEC 60529 and NEMA 250: ▪ IP66, NEMA Type 4X ▪ IP68, NEMA Type 6P (24 h at 1.83 m (6.00 ft) under water)
Vibration resistance	▪ Stochastic noise (random sweep) as per IEC 60068-2-64 Case 2 ▪ Guaranteed for 5 to 2 000 Hz: 1.25 $(m/s^2)^2/Hz$, ~ 5 g
Electromagnetic compatibility (EMC)	▪ Electromagnetic compatibility as per EN 61326 series and NAMUR recommendation EMC (NE21) ▪ Maximum measured error during EMC testing: < 0.5 % of the span. For more details, refer to the EU Declaration of Conformity (www.endress.com/downloads).

Process

Process temperature, process pressure

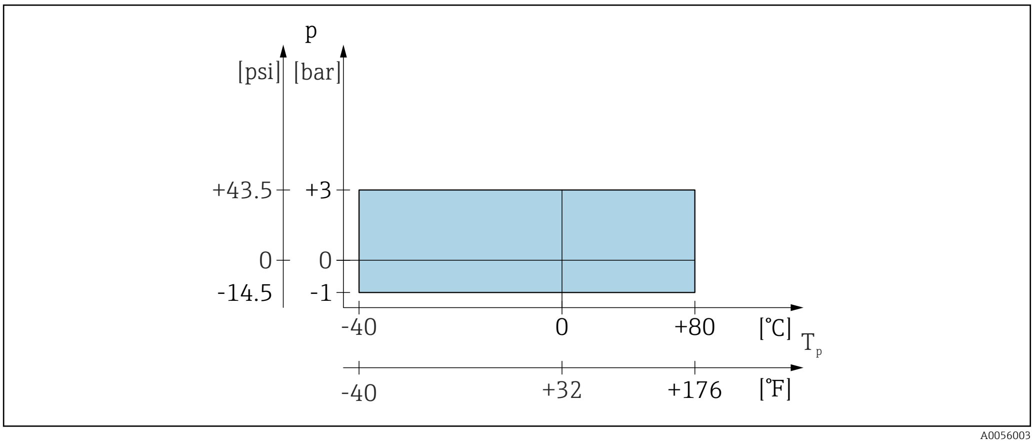
The maximum pressure for the device depends on the lowest-rated element with regard to pressure.

Components are: process connection, optional mounting parts, or accessories.

⚠ WARNING

Incorrect design or use of the device may cause injury due to bursting parts!

- ▶ Only operate the device within the specified limits for the components!
- ▶ MWP (Maximum Working Pressure): The MWP is specified on the nameplate. This value refers to a reference temperature of +20 °C (+68 °F) and may be applied to the device for an unlimited time. Observe the temperature dependency of the maximum working pressure. For flanges, refer to the following standards for the permitted pressure values at higher temperatures: EN 1092-1 (with regard to their stability/temperature property, the materials 1.4435 and 1.4404 are grouped together under EN 1092-1; the chemical composition of the two materials can be identical), ASME B16.5, JIS B2220 (the latest version of the standard applies in each case). Maximum working pressure data that deviate from this are provided in the relevant sections of the Technical Information.
- ▶ The Pressure Equipment Directive (2014/68/EU) uses the abbreviation **PS** . This corresponds to the maximum working pressure (MWP) of the device.



20 Permitted range for process temperature and process pressure

Process temperature range
−40 to +80 °C (−40 to +176 °F)

Process pressure range, 40 mm (1.5 in) antenna
- p_{gauge} = −1 to 3 bar (−14.5 to 43.5 psi)
- p_{abs} < =4 bar (58 psi)

Process pressure range, 80 mm (3 in) antenna with UNI slip-on flange 3", 4"
- p_{gauge} = −1 to 1 bar (−14.5 to 14.5 psi)
- p_{abs} < =2 bar (29 psi)

Process pressure range, 80 mm (3 in) antenna with UNI slip-on flange 6"
For unpressurized applications

The pressure range may be further restricted in the case of a CRN approval.

Relative permittivity

For liquids
- $\varepsilon_r \geq 1.8$
- Contact Endress+Hauser for lower ε_r values

For bulk solids
$\varepsilon_r \geq 1.6$

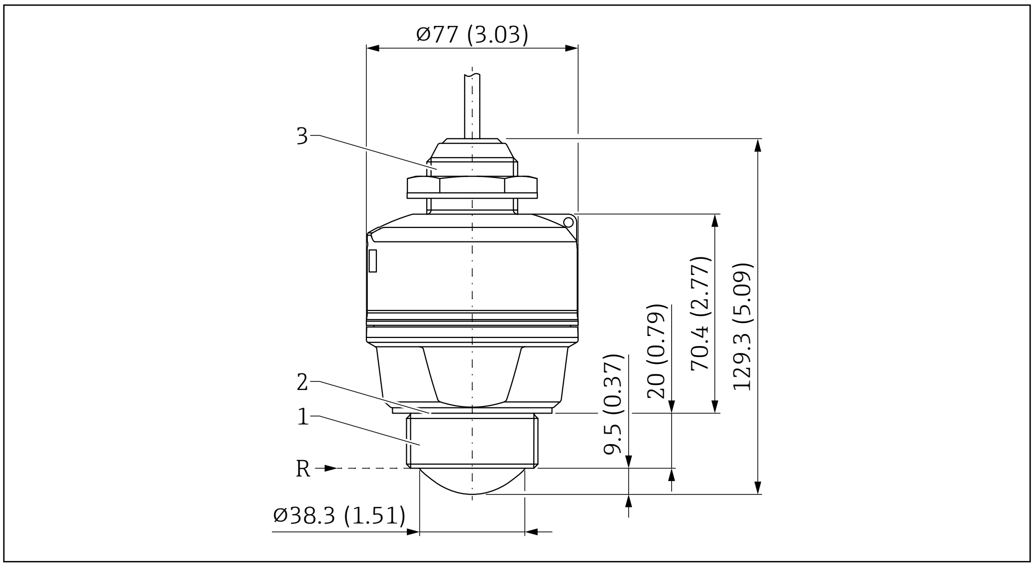
For applications with a lower relative permittivity than indicated, contact Endress+Hauser.

For the relative permittivity values (ε_r values) of many media commonly used in industry, please refer to:
- Relative permittivity (ε_r value), Compendium CP01076F
- The Endress+Hauser "DC Values app" (available for Android and iOS)

Mechanical construction

Dimensions

40 mm (1.5 in) antenna, cable entry from above

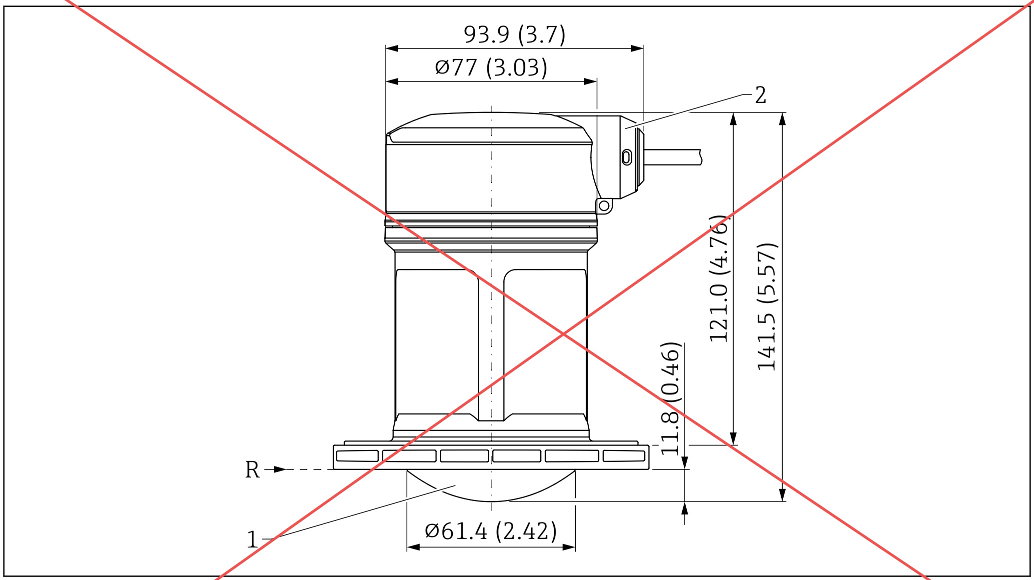


21 Dimensions; 40 mm (1.5 in) antenna with cable entry from above. Unit of measurement mm (in)

R Reference point of the measurement
1 Antenna end process connection, thread
2 EPDM seal (G 1 1/2 thread)
3 Cable entry from above process connection

The seal thickness is 2 mm (0.08 in).

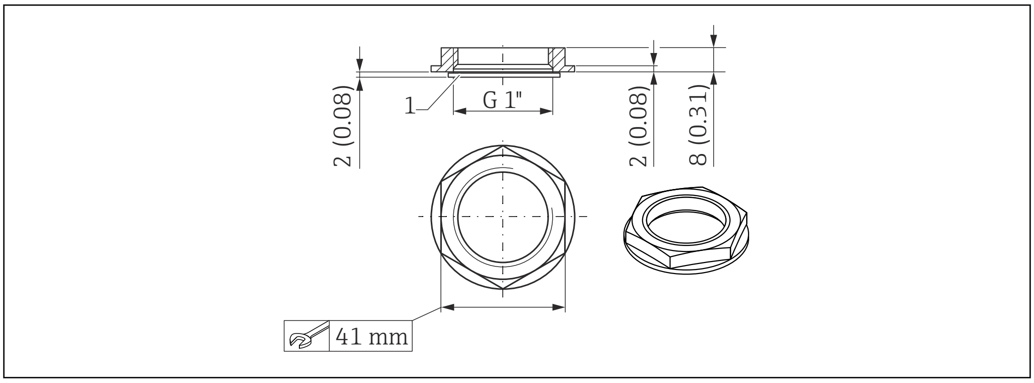
~~80 mm (3 in) antenna, cable entry at the side~~



~~24 Dimensions; 80 mm (3 in) antenna with cable entry at the side~~

~~R Reference point of the measurement~~
~~1 Antenna end process connection, without; prepared for UNI slip-on flange~~
~~2 Side cable entry~~

Counter nut, cable entry from above process connection



25 *Dimensions; counter nut, cable entry from above process connection. Unit of measurement mm (in)*

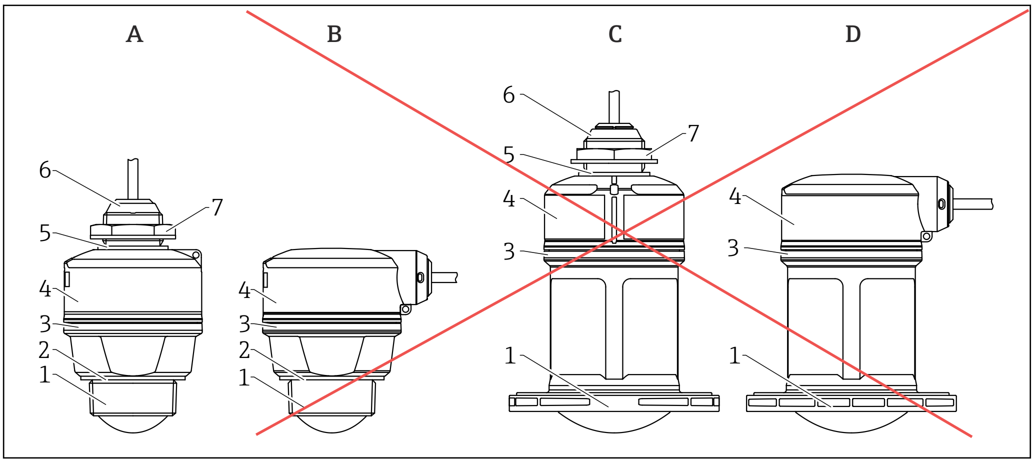
1 Seal

- The counter nut with seal (EPDM) is included in the scope of delivery
- Material: PA6.6

Weight

Weight (including 5 m (16 ft) cable)
- Device with 40 mm (1.5 in) antenna: approx. 0.5 kg (1.1 lb)
- Device with 80 mm (3 in) antenna: approx. 0.7 kg (1.5 lb)

Materials



A0055121

26 *Device design*

A 40 mm (1.5 in) antenna, cable entry from above
B 40 mm (1.5 in) antenna, cable entry at the side
C 80 mm (3 in) antenna, cable entry from above
D 80 mm (3 in) antenna, cable entry at the side
1 Antenna end process connection; PVDF
2 EPDM seal (for G 1½" thread)
3 PBT/PC design ring
4 Sensor housing/Cable entry process connection; PBT/PC (for dust ignition-proof devices: PC)
5 EPDM seal
6 Cable entry; PBT/PC (for dust ignition-proof devices: PC)
7 Counter nut; PA6.6

Connecting cable

Available cable length: 5 to 300 m (16 to 980 ft)

Material : PVC

For devices with marine approval: halogen-free cable (material: XLPE = connected polyethylene)

Operability

Operating concept

Operator-oriented menu structure for user-specific tasks
- Guidance
- Diagnostics
- Application
- System

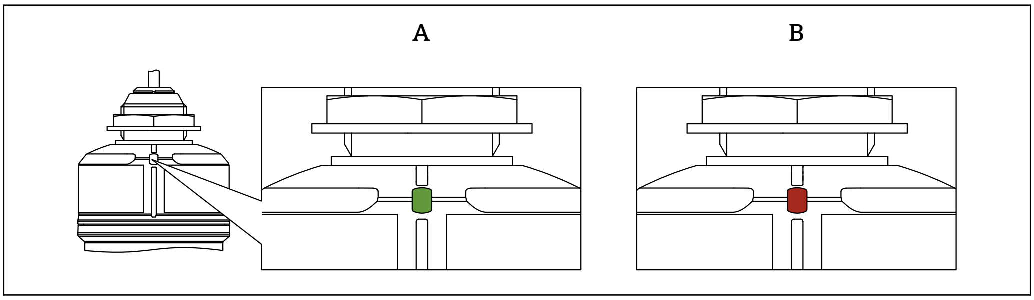
Fast and safe commissioning
- Interactive wizard with graphical interface for guided commissioning in FieldCare/DeviceCare or SmartBlue app
- Menu guidance with brief descriptions of the individual parameter functions

Integrated data memory
Up to 100 event messages recorded in the device

Efficient diagnostic behavior increases measurement reliability
- Remedial action is integrated in plain text
- Diverse simulation options

Bluetooth® wireless technology (optional)
- Quick and easy setup with the SmartBlue app or Field Xpert SMT70/SMT77
- No additional tools or adapters required
- Encrypted single point-to-point data transmission (tested by Fraunhofer Institute) and password-protected communication via Bluetooth® wireless technology
- The device can be retrofitted with Bluetooth® wireless technology

LED indicator



A0055146

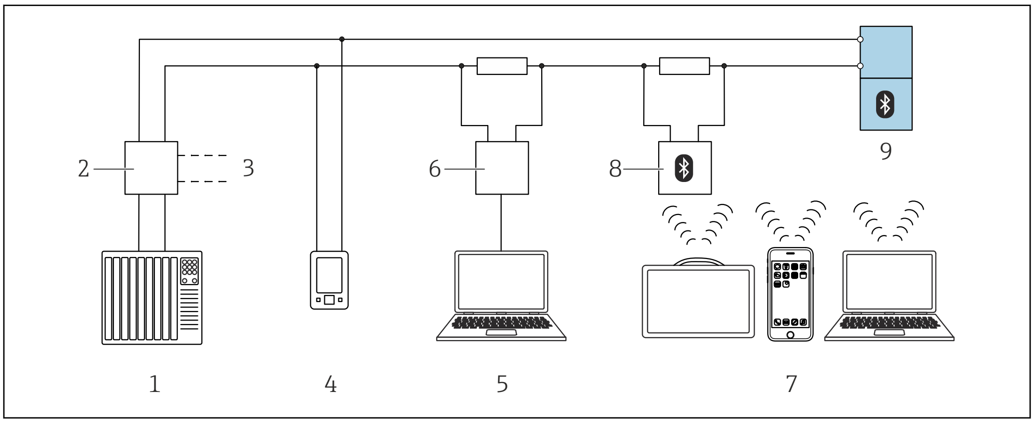
27 *LED indicator on the device*

A Green LED
B Red LED

Functions:
- Displays the operating status
  - Operation (green)
  - Fault (red)
- Displays an active Bluetooth® connection (flashing)

Remote operation

Via HART protocol or Bluetooth® wireless technology



A0044334

28 *Options for remote operation via HART protocol*

1 PLC (programmable logic controller)
2 Transmitter power supply unit, e.g. RN42 (with communication resistor)
3 Connection for Commubox FXA195 and AMS Trex™ device communicator
4 AMS Trex™ device communicator
5 Computer with operating tool (e.g. DeviceCare/FieldCare , AMS Device View, SIMATIC PDM)
6 Commubox FXA195 (USB)
7 Field Xpert SMT70/SMT77, smartphone or computer with operating tool (e.g. DeviceCare)
8 Bluetooth® modem with connecting cable (e.g. VIATOR)
9 Transmitter

Operation via Bluetooth® wireless technology (optional)

Prerequisite
- Device with Bluetooth® wireless technology order option
- Smartphone or tablet with Endress+Hauser SmartBlue app or PC with DeviceCare from version 1.07.07 or Field Xpert SMT70/SMT77

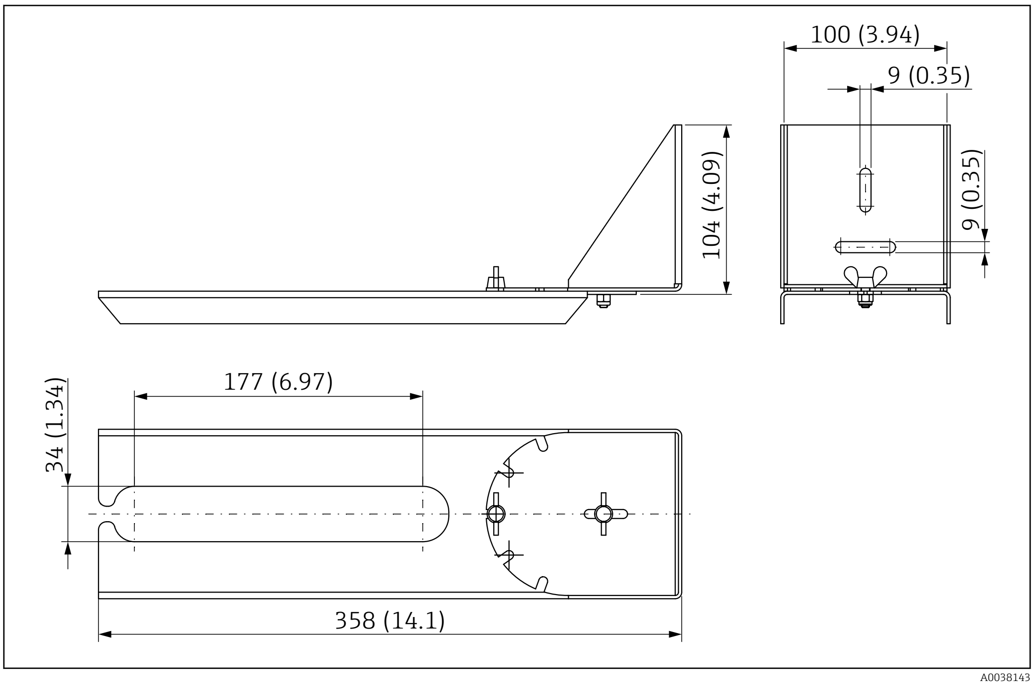
The connection has a range of up to 25 m (82 ft). The range can vary depending on environmental conditions such as attachments, walls or ceilings.

Supported operating tools

Smartphone or tablet with Endress+Hauser SmartBlue app, DeviceCare from version 1.07.07, FieldCare, AMS and PDM

Pivotable mounting bracket

The pivotable mounting holder is used, for example, to install the device in a manhole over a sewer channel.



51 *Dimensions of pivotable mounting bracket. Unit of measurement mm (in)*

34 mm (1.34 in) openings for all G 1" or MNPT 1" cable entry thread process connections

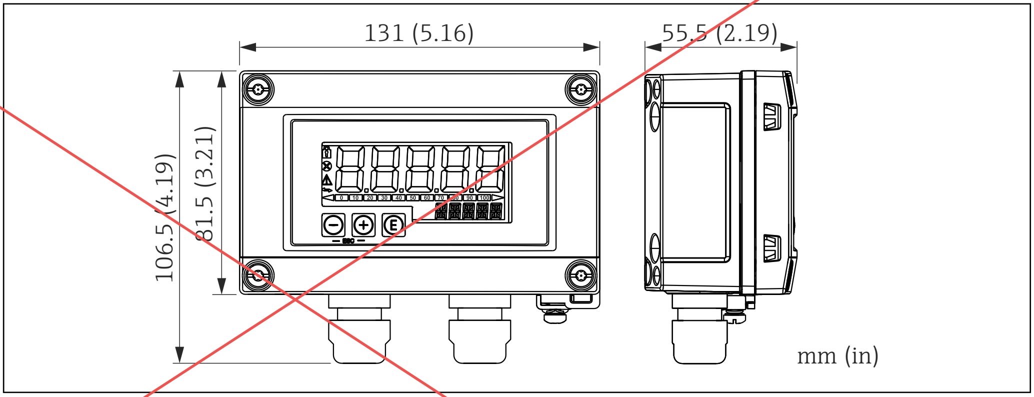
Material
316L (1.4404)

Order code
71429910

FAU40 alignment unit

The alignment unit is used to optimally align the sensor with the bulk solids.

RIA15 in the field housing



A0017722

60 *Dimensions of RIA15 in the field housing. Unit of measurement mm (in)*

Older device versions before February 2025 can only be connected via the 4 to 20 mA current output.

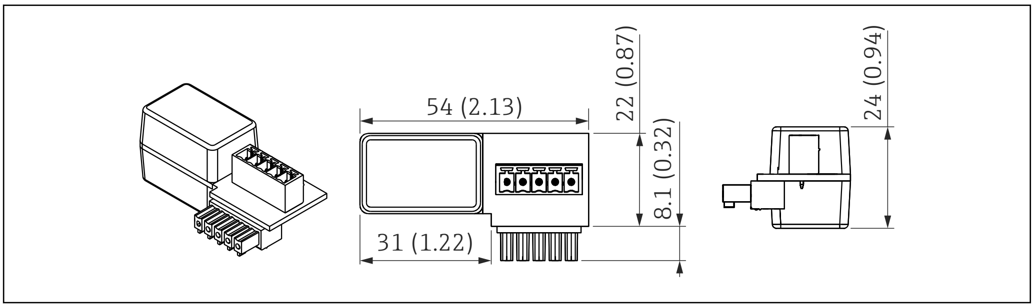
The remote display RIA15 can be ordered with or without operation via the product structure "Accessory enclosed".

Field housing material: Plastic (PBT with steel fibers, antistatic)

Other housing versions are available via the RIA15 product structure.

Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

HART communication resistor

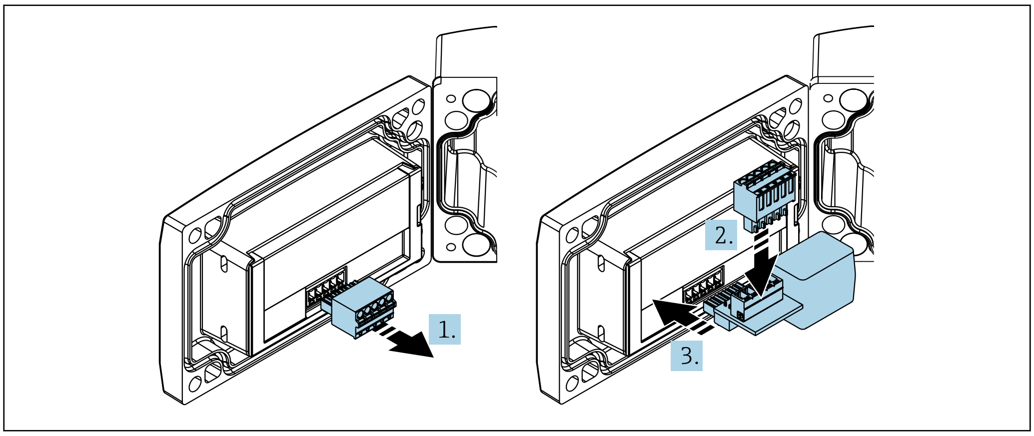


A0020858

61 *Dimensions of HART communication resistor. Unit of measurement mm (in)*

The HART communication resistor is required for operation of the RIA15 and is supplied with "remote display RIA15, with operation via HART" when ordering.

Document Technical Information TI01043K and Operating Instructions BA01170K



A0020844

In order to operate the RIA15, the HART communication resistor must be integrated.

1. Disconnect the plug-in terminal block.
2. Insert the terminal block into the slot provided on the HART communication resistor module.
3. Insert the HART communication resistor in the slot in the housing.

DeviceCare SFE100

Configuration tool for IO-Link, HART, PROFIBUS and FOUNDATION Fieldbusfield devices
DeviceCare is available for download free of charge at www.software-products.endress.com. You need to register in the Endress+Hauser software portal to download the application.

Technical Information TI01134S

FieldCare SFE500

FDT-based plant asset management tool
It can configure all intelligent field units in your system and helps you manage them. By using the status information, it is also a simple but effective way of checking their status and condition.

Technical Information TI00028S

Device Viewer

All the spare parts for the device, along with the order code, are listed in the *Device Viewer* (www.endress.com/deviceviewer).

Commubox FXA195 HART

For intrinsically safe HART communication with FieldCare via the USB interface

Technical Information TI00404F

RN22

Single or two-channel active barrier for safe electrical isolation of 4 to 20 mA standard signal circuits, HART transparent

Technical Information TI01515K and Operating Instructions BA02004K

RN42

Single-channel active barrier with wide-range power supply for safe electrical isolation of 4 to 20 mA standard signal circuits, HART transparent.

Technical Information TI01584K and Operating Instructions BA02090K

Field Xpert SMT70

Universal, high-performance tablet PC for device configuration in Ex Zone 2 and non-Ex areas

For details, see "Technical Information" TI01342S

Field Xpert SMT77

Universal, high-performance tablet PC for device configuration in Ex Zone 1 areas

For details, see "Technical Information" TI01418S

SmartBlue app	Mobile app for easy configuration of devices on site via Bluetooth® wireless technology.
RMA42	Digital process transmitter for monitoring and displaying analog measured values For details, see Technical Information TI00150R and Operating Instructions BA00287R

Documentation

For an overview of the scope of the associated Technical Documentation, refer to the following:
- *Device Viewer* (www.endress.com/deviceviewer): Enter the serial number from the nameplate
- *Endress+Hauser Operations app*: Enter serial number from nameplate or scan matrix code on nameplate.

Standard documentation

Document type: Operating Instructions (BA)
Installation and initial commissioning – contains all the functions in the operating menu that are needed for a routine measuring task. Functions beyond this scope are not included.

Document type: Description of Device Parameters (GP)
The document is part of the Operating Instructions and serves as a reference for parameters, providing a detailed explanation of each individual parameter of the operating menu.

Document type: Brief Operating Instructions (KA)
Quick guide to the first measured value – includes all essential information from incoming acceptance to electrical connection.

Document type: Safety Instructions, certificates
Depending on the approval, safety instructions are supplied with the device, e.g. XA. This documentation is an integral part of the Operating Instructions.
Information on the Safety Instructions (XA) that are relevant for the device is provided on the nameplate.

Supplementary device-dependent documentation

Additional documents are supplied depending on the device version ordered: Always comply strictly with the instructions in the supplementary documentation. The supplementary documentation is an integral part of the device documentation.

Registered trademarks

Apple®
Apple, the Apple logo, iPhone, and iPod touch are trademarks of Apple Inc., registered in the U.S. and other countries. App Store is a service mark of Apple Inc.

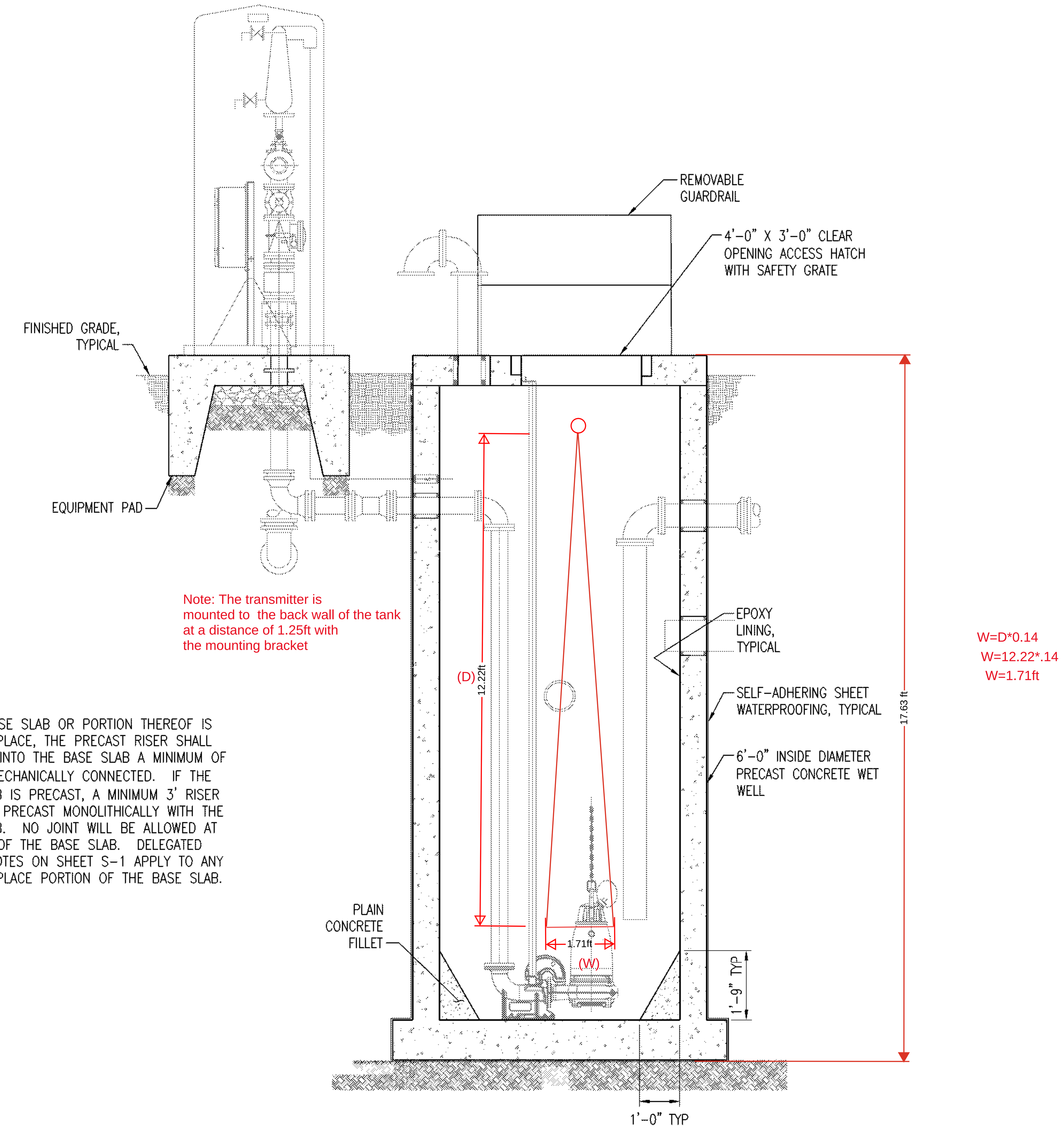
Android®
Android, Google Play and the Google Play logo are trademarks of Google Inc.

Bluetooth®
The *Bluetooth*® word mark and logos are registered trademarks owned by the Bluetooth SIG, Inc. and any use of such marks by Endress+Hauser is under license. Other trademarks and trade names are those of their respective owners.

HART®
Registered trademark of the FieldComm Group, Austin, Texas USA

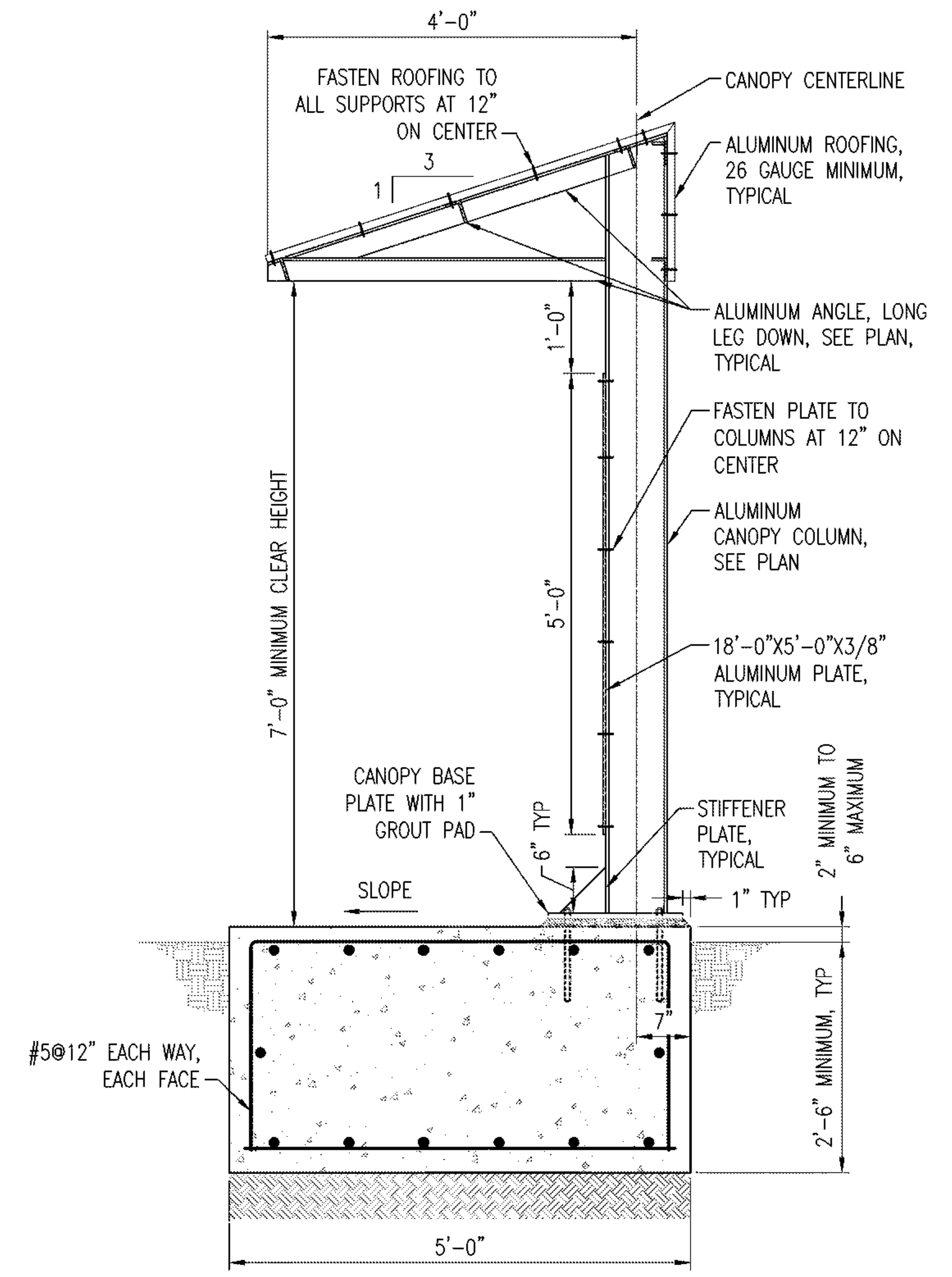
GENERAL SHEET NOTES

1. REFER TO SHEET S-1 FOR GENERAL STRUCTURAL NOTES.
2. REFER TO SHEET S-2 FOR STRUCTURAL TYPICAL DETAILS.
3. COORDINATE WITH THE MECHANICAL AND ELECTRICAL DRAWINGS FOR THE SIZE AND LOCATIONS OF SLAB AND WALL PENETRATIONS, AND FOR THE LOCATIONS OF EQUIPMENT PADS.
4. WALL SLEEVES FOR PIPING, ACCESS HATCH FRAME AND OTHER INSERTS MUST BE CAST INTO THE STRUCTURE AT THE PLACE OF MANUFACTURE.
5. DETERMINE AND SUBMIT FOR APPROVAL EXACT LOCATIONS AND QUANTITIES OF PIPE SUPPORTS REQUIRED.



NOTE:
IF THE BASE SLAB OR PORTION THEREOF IS CAST-IN-PLACE, THE PRECAST RISER SHALL BE INSET INTO THE BASE SLAB A MINIMUM OF 8" AND MECHANICALLY CONNECTED. IF THE BASE SLAB IS PRECAST, A MINIMUM 3' RISER SHALL BE PRECAST MONOLITHICALLY WITH THE BASE SLAB. NO JOINT WILL BE ALLOWED AT THE TOP OF THE BASE SLAB. DELEGATED DESIGN NOTES ON SHEET S-1 APPLY TO ANY CAST-IN-PLACE PORTION OF THE BASE SLAB.

A / S-4 SECTION
1/2" = 1'-0"
REF: S-3



NOTES:
1. ALL CANOPY FRAMING MUST BE ALUMINUM.
2. FASTEN PLATES TO FRAMING WITH TYPE 316 STAINLESS STEEL SELF DRILLING SCREWS.
3. PROVIDE FULLY WELDED CONNECTIONS AT ALL CANOPY FRAMING CONNECTIONS WITH 3/16" MINIMUM ALL AROUND FILLET WELDS UNLESS NOTED OTHERWISE, COPE ANGLES AS NECESSARY FOR FIT UP.
4. SLOPE THE TOP OF THE CONCRETE SLAB AWAY FROM THE CANOPY CENTERLINE AT 1/4" PER 1'-0".

B / S-4 SECTION
3/4" = 1'-0"
REF: S-3

GRAPHIC SCALE
0 1' 2' 3'
SCALE: 3/4" = 1'-0"
0 1' 2' 4'
SCALE: 1/2" = 1'-0"

NO.	CONSTRUCTION REVISION	DATE

PROJECT NO. UN-1219

PROFESSIONAL CERTIFICATION. I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND. LICENSE NO. 42863, EXPIRATION DATE: 12/2/2024.

Whitman, Requardt & Associates, LLP
801 South Caroline Street, Baltimore, Maryland 21231

OWNER:
CHARLES COUNTY GOVERNMENT, DPW
5310 HAWTHORNE ROAD
LA PLATA, MARYLAND 20646
301-609-7400 301-753-8270

PROJECT NO. UN-1219

POMONKEY PUMP STATION IMPROVEMENTS
STRUCTURAL SECTIONS AND DETAILS

CHARLES COUNTY, MARYLAND
ELECTION DISTRICT NO. 6

SCALE	SHEET	DRAWING	WO:14475-006
AS SHOWN	S-4	NO. 12 OF 26	MAY 2024

POMONKEY PUMP STATION IMPROVEMENTS

N:\14475-006\CADD\14475065-04.dwg May 09, 2024 - 9:05am

Data Sheet 2

Customer: Reference: Date:	Johnston Construction 32384 Charles County Pomonkey Pump Station Improvements 04/16/2026
<u>Qty</u>	<u>Description</u>
6	Mfg: Anchor Scientific: Float Switch, NO Model Number: GSE60NONC-GOLD Tags/Service: LSLL-01, LSL-01, LSH-01, LSHH-01, LSL-03, LSH-03 / Pump Station Level Float Switches Specifications: • G - Model G float switch • SE - External weight mounting style • 60 - 60ft cable length • NONC - 1 normally open, 1 normally closed contact • GOLD - For use in intrinsically safe applications

FLOAT SWITCHES

TAGS: LSLL-01, LSL-01, LSL-02, LSH-03, LSHH-01, LSH-02

Float Level Switches

UL Listed

anchor scientific inc.

roto-float TYPE S - SUSPENDED & TYPE P - PIPE MOUNTED

Priced From $47.00

Priced From $47.00

TYPES - SUSPENDED & TYPE P - PIPE MOUNTED

The ROTO-FLOAT is a direct acting float switch. Each ROTO-FLOAT contains a single pole mercury switch which actuates when the longitudinal axis of the float is horizontal, and deactuates when the liquid level falls 1" below the actuation elevation.

The float is a chemical resistant polypropylene casing with a firmly bonded electrical cable protruding. One end of the cable is permanently connected to the enclosed mercury switch and the entire assembly is encapsulated to form a completely watertight and impact resistant unit. Type S - Suspended has built-in weight.

ROTO-FLOATS can be mounted on a support pipe (type P) or suspended from above (type S). Advantages of the ROTO-FLOAT are low cost, simplicity and reliability.

NOTE: Mercury switches are not to be used in potable water.

MATERIALS OF CONSTRUCTION
Float housing........Polypropylene
Cable clamp..........Polypropylene
Cable jacket..........P.V.C

Applications

- Pilot Duty
- Industrial Control Equipment

CABLE
P.V.C. type STO #18 conductors (41 strand) rated 600 volts

- Various lengths available
- See table of models
- Non-standard lengths also available on special order.

Ordering information

Switch Arrangement No.	Cable Length	Suspended Type S Model No .	Pipe Mounted Type P Model
Normally Open	20 30 40	S20NO S30NO S40NO	P20NO P30NO P40NO
Normally	20 30	S20NC S30NC	P20NC P30NC

eco-float Model G

Applications

The Eco-Float can be used in a variety of liquid level monitoring applications including sumps, sewage ejectors, septic tanks, vaults, lift stations, and tanks. Eco-Floats are ruggedly constructed of corrosion resistant materials, enabling them to be used in a variety of different liquids. Some applications are subject to additional requirements described in the National Electric Code.

Priced From $27.50

Description

The Eco-float is a **mercury-free** float switch for controlling liquid levels in a variety of applications. A snap-action switch is activated by a steel ball rolling back and forth within a switching tube in a plastic float housing. There is a minimum differential between “on” and “off” of approximately 3.5 inches. Greater differentials can be achieved when the pipe mounted or externally

FEATURES

- **Mercury Free**
- **Variety of Mounting Styles**
- **Variety of Circuit Configurations**
- **Installation Easy**
- **Differential In One Float**
- **Replaces Diaphragm and Mercury Switches**

Ordering information

60 - 60ft cable length

Model	Mounting Style	Cable Length*	Circuit Configuration
G	SE (external weight) ~~SI (internal weight)~~ ~~P (pipe mounted)~~	~~10”~~ ~~15”~~ ~~20”~~ ~~30”~~ ~~40”~~ ~~50”~~	~~NO (normally open)~~ ~~NC (normally closed)~~ NONC (normally open/normally closed)

* Custom lengths available.
Example: GSI 20NO - Eco-Float, suspended internal wt. 20’, normally opens contacts.

-GOLD FOR INTRINSICALLY SAFE APPLICATION

solo-float

DESCRIPTION

The new Solo-Float" is a direct acting **mercury-free** float switch for controlling 11/2 HP and smaller pump motors. The new Solo-Float has been completely redesigned and uses a metal ball rolling back and forth in a tube to actuate a reliable, snap-action switch. An adjustable, plastic clamp with stainless steel hose clamp is provided for attaching the Solo-Float at the appropriate level on the pump discharge pipe. By varying the tether length, a pumping range of 8" - 14" can be achieved. The Solo-Float is available with a piggyback plug in either a 115 volt or 230 volt configuration. It is also available blunt cut, without a plug, for wiring to an Anchor Scientific JX junction box.

Priced From $28.00

FEATURES

- **Reliability**
- **Adjustable Pumping Range**
- **Manual Operation**
- **Non-mercury Element**
- **Converts Nonautomatic Pumps To Automatic**
- **Replaces Diaphragm Pressure Switches**
- **Easy Installation**
- **In Stock, Low Cost**

Ordering information

Model	Cable Length*	Circuit Configuration	Plug Voltage
D	10’ 15’ 20’ 25’ 30’	NO (normally open) NC (normally closed)	115 for 115 volt 230 for 230 volt

* Custom lengths available.

Example:

Model	Cable Length*	Circuit Configuration	Plug Voltage
D	20’	NO (normally open)	115 for 115 volt

Data Sheet 4

Customer: Reference: Date:	Johnston Construction 32384 Charles County Pomonkey Pump Station Improvements 04/16/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: E+H: Digital Process Meter Display Model Number: RIA452-C211A11A Tags/Service: LEVEL DISPLAY / PLC/RTU Panel Process Meter Display Specifications: • RIA452 - Process indicator with pump control unit • C - FM AIS, I, II, III/2/ABCDEFG approvals • 2 - 20-30VDC, 20-28VAC power supply • 1 - 0/4-20mA measuring signal • 1 - 4 output relays, SPDT • A - Standard communication • 1 - 96x96 panel mounting housing, front IP65 • 1 - Basic version • A - Standard version

TI00113R/09/EN/14.19
71468732
2019-12-30

DIGITAL PROCESS METER DISPLAY

TAGS: LEVEL DISPLAY

Technical Information
RIA452

Process indicator

Digital process indicator in panel mounted housing for monitoring and displaying analog measured values with pump control and batch functions and flow calculation

Application

- Water/wastewater sector
- ~~Power industry~~
- ~~Raw materials~~
- ~~Chemicals industry~~
- ~~Food industry~~

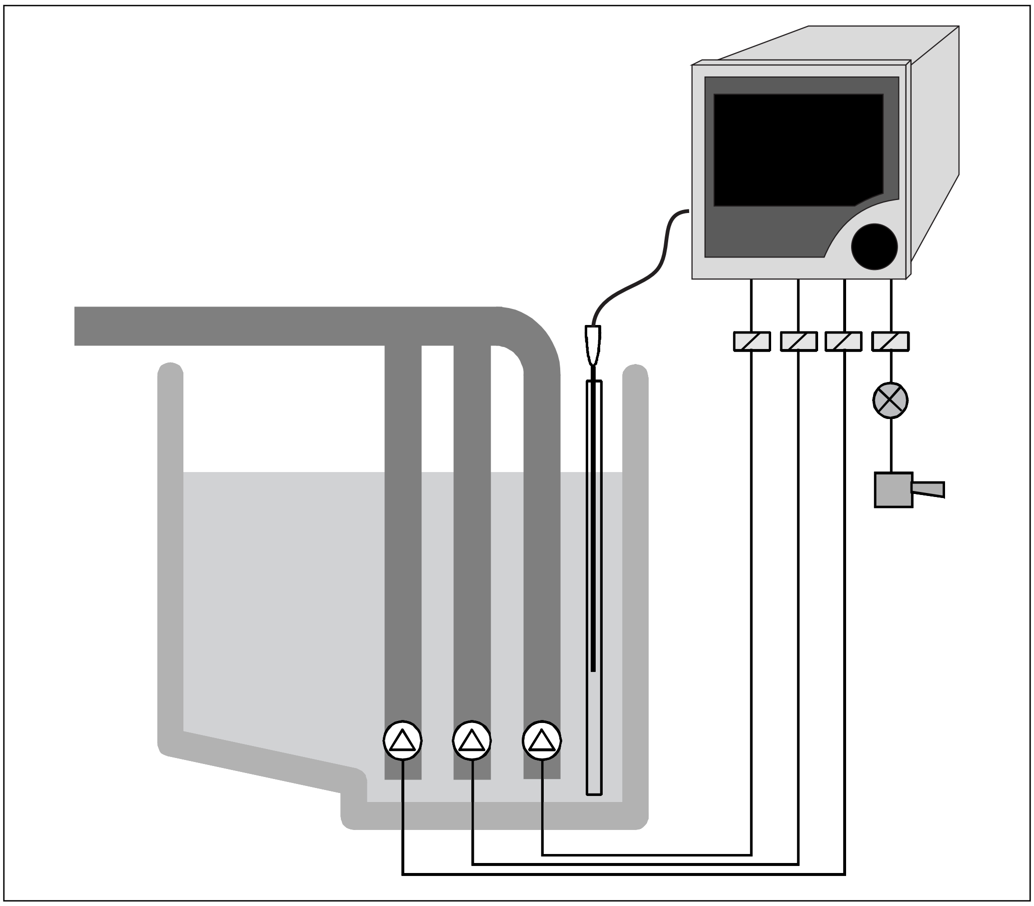
Your benefits

- 7-digit 14-segment LC display
- Multicolored
- Large bar graph with overrange and underrange
- Intrinsically safe input with transmitter power supply
- Digital status inputs for pump monitoring
- Universal input
- Up to eight relays
- Min/Max value saved
- Pump control functions
- Batch functions
- Flow measurement for open channels and weirs
- Linearization table with 32 support points
- Analog output
- Pulse output with totalizer
- Jog-shuttle operation
- Freely programmable units
- Configuration via interface and operating software
- Tank linearization via PC software

Endress+Hauser
People for Process Automation

Function and system design

Measuring principle



A0028466

1 *Example of an application of the process indicator*

The single-channel RIA452 process indicator monitors and displays analog measured values. Pumps can be monitored with the digital status inputs. The measured value is displayed using the seven-digit, 14-segment LC display. Numbers and units are displayed in white, the bar graph in yellow, overrange and underrange in red, and the limit value flags and digital status inputs in green and yellow. The RIA452 can provide power directly to two-wire transmitters that are connected. As an option, it is also possible to select the input and the transmitter power supply as intrinsically safe for Ex applications. Up to eight freely programmable relays monitor the measured value for limit value overshoot and undershoot. Other operating modes for the relays include sensor or device malfunction, batch and pump control functions (e.g. alternating pump control). Furthermore, the RIA452 can be used as a preset counter and for measuring flow at open channels and measuring weirs.

The scalable analog output offers many different ways of forwarding the input signal: zoom function, linearization, offset, inversion and signal conversion (input/output conversion). The optional pulse output allows users to output integrated process values.

Measuring system

Microcontroller-controlled indicator in a panel-mounted housing with a multicolored, illuminated LC display. Analog measured value acquisition is via an analog/digital converter. The digital status inputs are scanned cyclically. Power can be supplied directly to two-wire sensors with the transmitter power supply which is integrated as standard. The current input is optionally available as an intrinsically safe version for Ex applications. In this case, the RIA452 has a second, intrinsically safe transmitter power supply.

The freely scalable analog output is output via digital/analog conversion. The digital pulse output is output directly.

Up to eight relays are available in the device for monitoring limit values, pump control and batch functions.

The device can be operated either onsite via the jog/shuttle dial, or via the PC with an operating software. Operation can be locked using the hardware key or software code.

Linearization

The following flow curves are programmed into the device for open channels and weirs:
- Khafagi-Venturi flume
- ISO Venturi flume
- BST [1] Venturi flume
- Parshall flume
- Palmer-Bowlus flume
- Rectangular weir
- Rectangular weir with constriction
- NFX [2] rectangular weir
- NFX [2] rectangular weir with constriction
- Trapezoidal weir
- Triangular weir
- BST [1] triangular weir
- NFX [2] triangular weir

User-configurable flow formula

$Q = C * (h^{\alpha} + \gamma * h^{\beta})$

The parameters α, β, γ and C can be entered freely.

Linearization function

Up to 32 user-definable linearization points are available in the device for the linearization of the input, e.g. for tank linearization.

The linearization table for standard tanks and customer-specific tanks can be generated with the ReadWin 2000 operating software.

Input

Measured variable

- Current (standard)
- Digital inputs (standard)
- Current/voltage, resistance, RTD assembly, thermocouples (universal input option)

Measuring range

Current input:

Current:
- 0/4 to 20 mA +10% overrange, 0 to 5 mA
- Short-circuit current: max. 150 mA
- Input impedance: ≤ 5 Ω
- Response time: ≤ 100 ms

Universal input:

Current:
- ~~0~~/4 to 20 mA + 10% overrange, 0 to 5 mA
- Short-circuit current: max. 100 mA
- Input impedance: ≤ 50 Ω

~~Voltage:~~
- ~~±150 mV, ±1 V, ±10 V, ±30 V, 0 to 100 mV, 0 to 200 mV, 0 to 1 V, 0 to 10 V~~
- ~~Input impedance: ≥ 100 kΩ~~

~~Resistance:~~
~~30 to 3 000 Ω in 3/4-wire technology~~

~~RTD assembly:~~
- ~~Pt100/500/1000, Cu50/100, Pt50 in 3/4-wire technology~~
- ~~Measuring current for Pt100/500/1000 = 0.25 mA~~

1) BST: British Standard
2) NFX: French standard NFX 10-311

~~Thermocouple types:~~
- ~~J, K, T, N, B, S, R as per IEC584~~
- ~~D, C as per ASTME998~~
- ~~U, L as per DIN43710/GOST~~
- ~~Response time: ≤ 100 ms~~

Digital input:

Digital input:
- Voltage level −3 to 5 V low, 12 to 30 V high (as per DIN19240)
- Input voltage max. 34.5 V
- Input current typ. 3 mA with overload and reverse polarity protection
- Sampling frequency max. 10 Hz

Galvanic isolation

Towards all other circuits

Output

Output signal

- Relay, transmitter power supply (standard)
- Current, voltage, pulse, intrinsically safe transmitter power supply (option)

Signal on alarm

No measured value visible on the LC display, no background illumination, no sensor power supply, no output signals, relays behave in safety-oriented manner.

Current/voltage output

Analog output range:
0/4 to 20 mA (active), 0 to 10 V (active)

Load:
- ≤ 600 Ω (current output)
- Max. output current 22 mA (voltage output)

Signal characteristics:
Freely scalable signal

Galvanic isolation towards all other circuits

Pulse output (open collector)

Pulse output (open collector):
- Frequency range to 2 kHz
- I_{max} = 200 mA
- U_{max} = 28 V
- $U_{low/max}$ = 2 V at 200 mA
- Pulse width = 0.04 to 2 000 ms

Relay output

Signal characteristics:
Binary, switches when the limit value is reached

Switch function: limit relay switches for the operating modes:
- Minimum/maximum safety
- Alternating pump control function
- Batch function
- Time control
- Window function
- Gradient
- Device malfunction
- Sensor malfunction

Switching threshold:
Freely programmable

Hysteresis:
0 to 99%

Signal source:
- Analog input signal
- Integrated value
- Digital input

Number:
4 in basic unit ~~(can be extended to 8 relays, option)~~

Electrical specifications:
- Relay type: changeover
- Relay switching capacity: 250 V_{AC} / 30 V_{DC}, 3 A
- Switch cycles: typically 10^5
- Switching frequency: max. 5 Hz
- Minimum switching load: 10 mA / 5 V_{DC}

Galvanic isolation towards all other circuits

Mixed assignment of low and extra-low voltage circuits is not permitted for neighboring relays.

Transmitter power supply

Transmitter power supply 1, terminal 81/82 (optionally intrinsically safe):

Electrical specifications:
- Output voltage: 24 V ±15%
- Output current: max. 22 mA (for U_{out} ≥ 16 V, sustained short-circuit proof)
- Impedance: ≤ 345 Ω

Transmitter power supply 2, terminal 91/92:

Electrical specifications:
- Output voltage: 24 V ±15%
- Output current: max. 250 mA (sustained short-circuit proof)

Transmitter power supply 1 and 2:

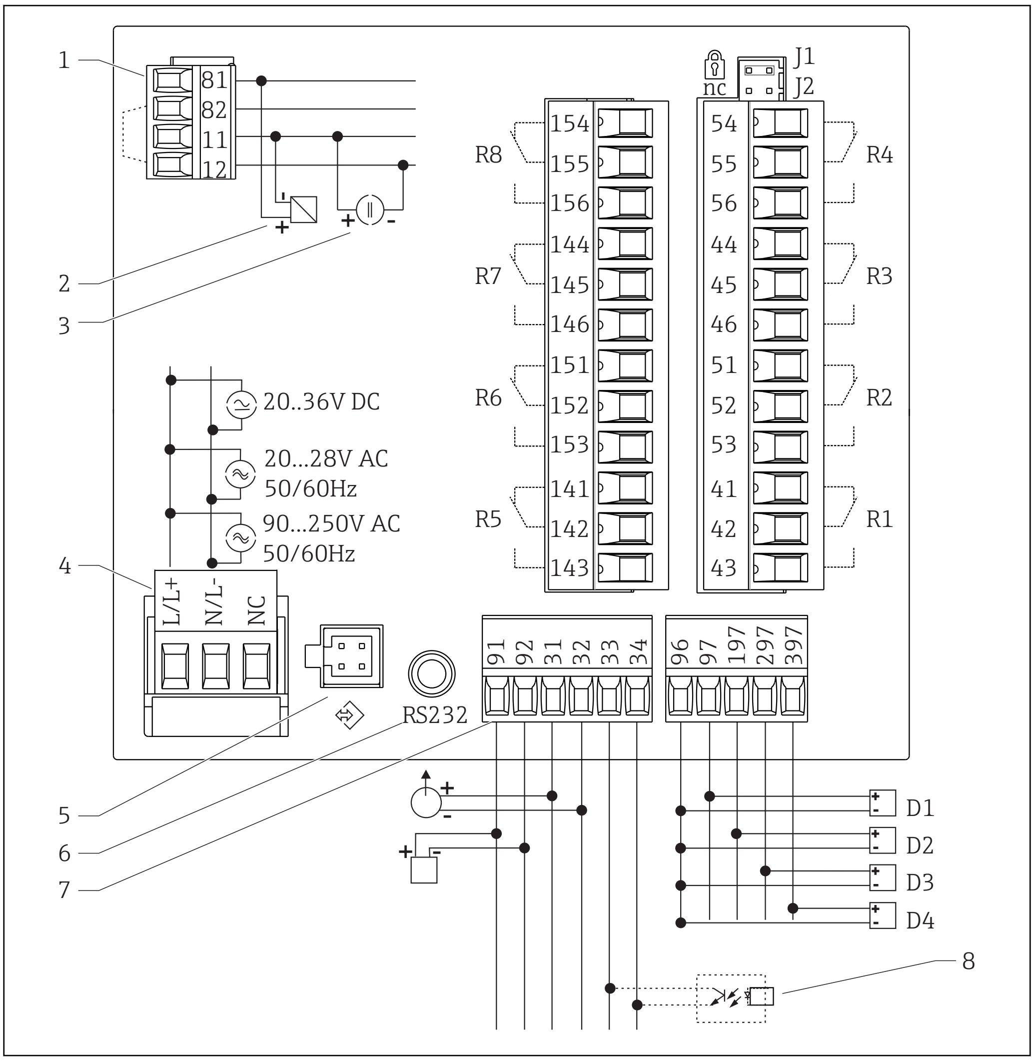
Galvanic isolation:
Towards all other circuits

HART®

HART® signals are not affected

Power supply

Terminal assignment

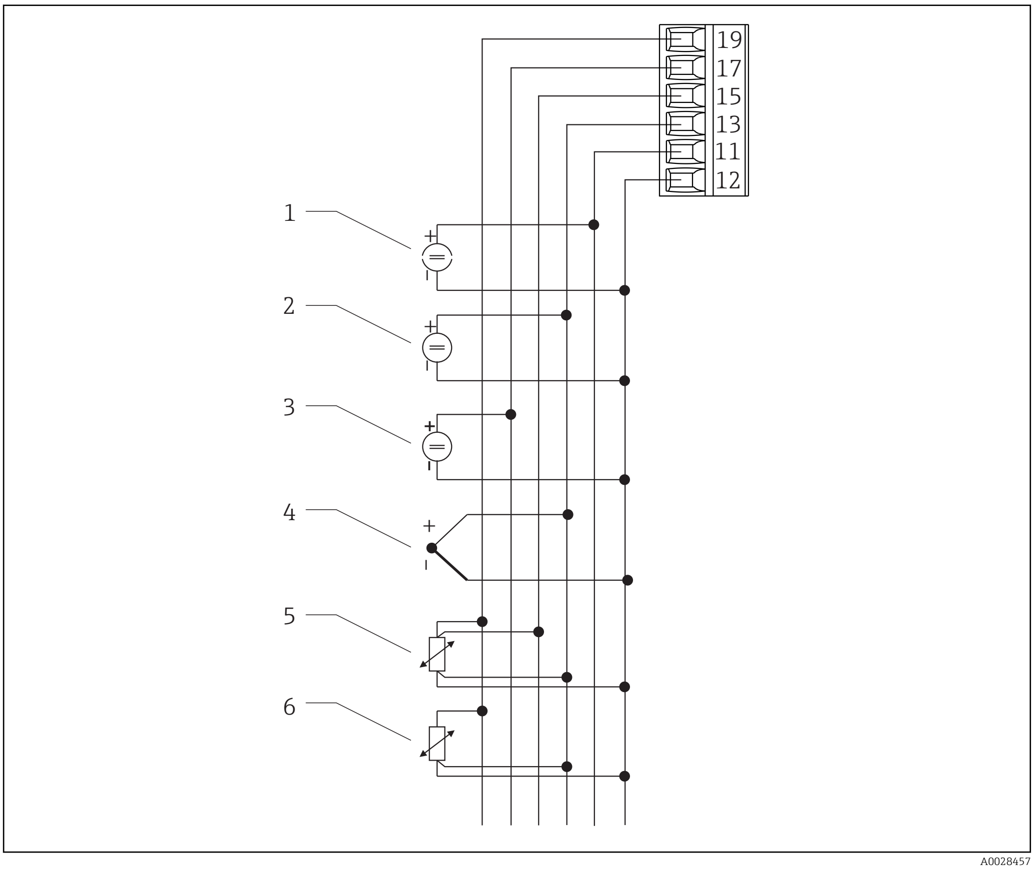


A0028439

2 *Terminal assignment of process indicator*

1	*Current input (12 and 82 jumpered internally)*
2	*- passive sensor*
3	*- active sensor*
4	*Power supply*
5	*Interface for PC operating software*
6	*RS232 interface*
7	*Transmitter power supply and analog output*
8	*Open collector output*
D1 to D4	*Digital inputs*
R1 to R4	*Relay outputs*
R5 to R8	*Relay outputs (optional)*
J1	Hardware write protection

Universal input option



3 *Universal input terminal assignment*

1	*Current input 0/4 to 20 mA*	4	*Thermocouples*
2	*Voltage input ±1 V*	5	*RTD assembly, 4-wire*
3	*Voltage input ±30 V*	6	*RTD assembly, 3-wire*

Interface connection data

RS232
- Connection: jack socket 3.5 mm, rear of device
- Transmission protocol: ReadWin 2000
- Transmission rate: 38400 Baud

Supply voltage

- Low voltage power unit 90 to 250 V_{AC}50/60 Hz
- Extra-low voltage power unit 20 to 36 V_{DC} or 20 to 28 V_{AC} 50/60 Hz
  The device must be powered only by a power unit that operates using a limited energy circuit in accordance with UL/EN/IEC 61010-1, Section 9.4 and the requirements in Table 18.

Power consumption

Power consumption max. 24 VA

Performance characteristics

Reference operating conditions

Power supply: 230 V_{AC} ±10%, 50 Hz ±0.5 Hz

Warm-up period: 90 min

Ambient temperature: 25 °C (77 °F)

Maximum measured error

Current input

Accuracy	0.1% of full scale
Resolution	13 bit
Temperature drift	≤ 0.4%/10 K (18 °F)

Universal input

	Input:	Range:	Maximum measured error of measuring range (oMR):
Accuracy	Current	0 to 20 mA, 0 to 5 mA, 4 to 20 mA; overrange: to 22 mA	±0.10%
	Voltage > 1 V	0 to 10 V, ±10 V, ±30 V	±0.10%
	Voltage ≤ 1 V	±1 V, 0 to 1 V, 0 to 200 mV, 0 to 100 mV, ±150 mV	±0.10%
	Resistance thermometer	Pt100, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604, GOST) Pt500, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604) Pt1000, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604)	4-wire: ± (0.10% oMR + 0.3 K (0.54 °F) 3-wire: ± (0.15% oMR + 0.8 K (1.44 °F))
		Cu100, −200 to 200 °C (−328 to 392 °F) (GOST) Cu50, −200 to 200 °C (−328 to 392 °F) (GOST) Pt50, −200 to 600 °C (−328 to 1 112 °F) (GOST)	4-wire: ± (0.20% oMR + 0.3 K (0.54 °F) 3-wire: ± (0.20% oMR + 0.8 K (1.44 °F))
	Resistance measurement	30 to 3 000 Ω	4-wire: ± (0.20% oMR + 0.3 K (0.54 °F) 3-wire: ± (0.20% oMR + 0.8 K (1.44 °F))
	Thermocouples	Typ J (Fe-CuNi), −210 to 999.9 °C (−346 to 1 382 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ K (NiCr-Ni), −200 to 1 372 °C (−328 to 2 502 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −130 °C (−234 °F)
		Typ T (Cu-CuNi), −270 to 400 °C (−454 to 752 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −200 °C (−328 °F)
		Typ N (NiCrSi-NiSi), −270 to 1 300 °C (−454 to 2 372 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ B (Pt30Rh-Pt6Rh), 0 to 1 820 °C (32 to 3 308 °F) (IEC584)	± (0.15% oMR + 1.5 K (2.7 °F)) from 600 °C (1 112 °F)
		Typ D (W3Re/W25Re), 0 to 2 315 °C (32 to 4 199 °F) (ASTME998)	± (0.15% oMR + 1.5 K (2.7 °F)) from 500 °C (932 °F)
		Typ C (W5Re/W26Re), 0 to 2 315 °C (32 to 4 199 °F) (ASTME998)	± (0.15% oMR + 1.5 K (2.7 °F)) from 500 °C (932 °F)
		Typ L (Fe-CuNi), −200 to 900 °C (−328 to 1 652 °F) (DIN43710, GOST)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ U (Cu-CuNi), −200 to 600 °C (−328 to 1 112 °F) (DIN43710)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ S (Pt10Rh-Pt), 0 to 1 768 °C (32 to 3 214 °F) (IEC584)	± (0.15% oMR + 3.5 K (6.3 °F)) for 0 to 100 °C (32 to 212 °F) ± (0.15% oMR + 1.5 K (2.7 °F)) for 100 to 1 768 °C (212 to 3 214 °F)
		Typ R (Pt13Rh-Pt), −50 to 1 768 °C (−58 to 3 214 °F) (IEC584)	± (0.15% oMR + 1.5 K (2.7 °F)) for 100 to 1 768 °C (212 to 3 214 °F)
Resolution		16 bit	
Temperature drift		Temperature drift: ≤ 0.1%/10 K (18 °F)	

Current output

Linearity	0.1% of full scale
Resolution	13 bit
Temperature drift	Temperature drift: ≤ 0.1%/10 K (18 °F)
Output Ripple	10 mV at 500 Ω for frequencies ≤ 50 kHz

Voltage output

Linearity	0.1% of full scale
Resolution	13 bit
Temperature drift	Temperature drift: ≤ 0.1%/10 K (18 °F)

Installation

Mounting location

Panel, cut-out 92 x 92 mm (3.62x3.62 in) (see 'Mechanical construction').

Orientation

Horizontal +/- 45° in every direction

Environment

Ambient temperature range

−20 to 60 °C (−4 to 140 °F)

Storage temperature

−30 to 70 °C (−22 to 158 °F)

Operating altitude

< 3 000 m (9 840 ft) over MSL

Climate class

To IEC 60654-1, Class B2

Degree of protection

Front IP 65 / NEMA 4

Device casing IP 20

Shock and vibration resistance

2 Hz (+3/-0) ... 13.2 Hz: ±1 mm (±0.04 in)

13.2 to 100 Hz: 0.7 g

Electromagnetic compatibility (EMC)

CE compliance

Electromagnetic compatibility in accordance with all the relevant requirements of the IEC/EN 61326 series and NAMUR Recommendation EMC (NE21). For details refer to the EU Declaration of Conformity.

Maximum measured error <1% of measuring range.

Interference immunity as per IEC/EN 61326 series, industrial requirements.

Interference emission as per IEC/EN 61326 series, Class A equipment.

Electrical protection class

IEC 60529 (IP code) / NEMA 250

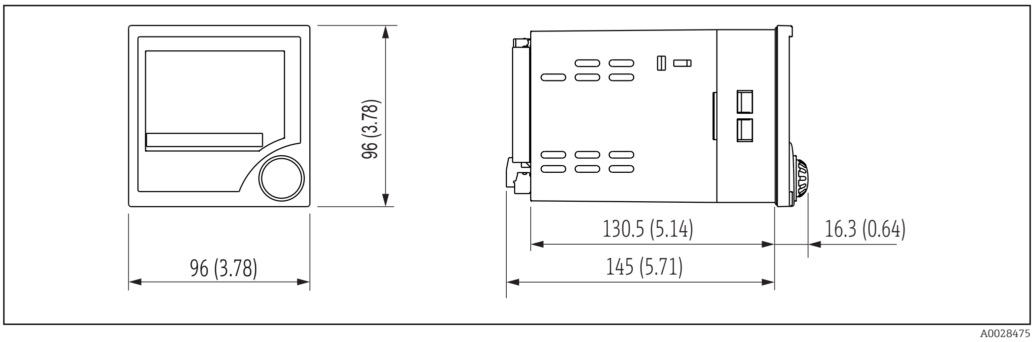
Condensation

Front: permitted

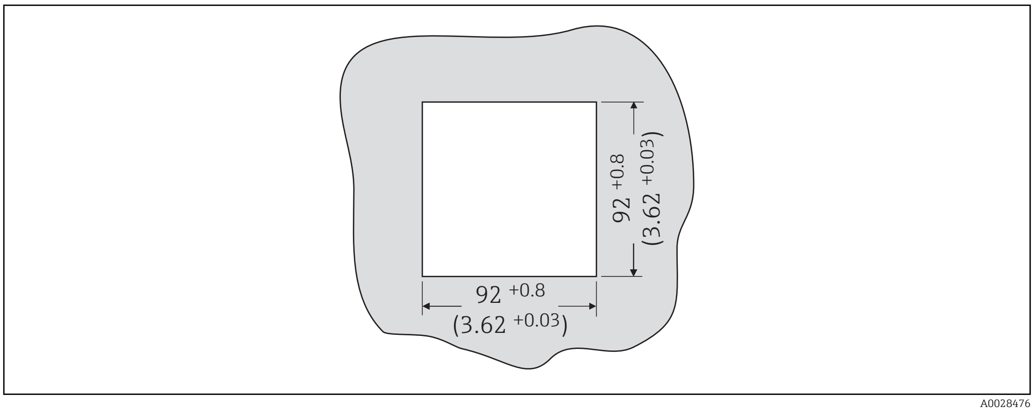
Device casing: not permitted

Mechanical construction

Design, dimensions



4 *Dimensions of the process indicator in mm (in)*



5 *Panel cutout, dimensions in mm (in)*

Weight

500 g (17.64 oz)

Materials

- Housing front: ABS plastic
- Housing casing: ABS GF plastic

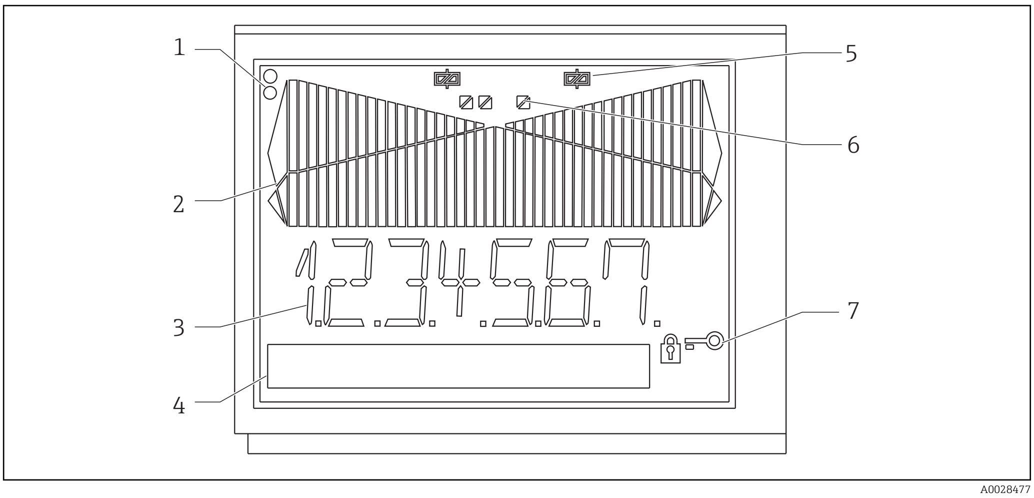
Terminals

Plug-in screw terminals, clamping range 1.5 mm^2 (16 AWG) solid, 1 mm^2 (18 AWG) strand with wire ferrule

Operability

Local operation

Display elements



6 Display elements of the process indicator

1 Device status LEDs: green - device ready for operation; red - device or sensor malfunction
2 Bar graph with overrange and underrange
3 7-digit 14-segment display
4 Unit and text field 9x77 dot matrix
5 Relay status indicator: if power is supplied to a relay, the symbol is displayed
6 Status indicator for digital inputs
7 Symbol for "device operation locked"

- Display range
  - -99999 to +99999 for measured values
  - 0 to 9999999 for counter values
- Signalization
  - Relay activation
  - Overrange/underrange

Operating elements

Jog/shuttle dial

Remote operation

Configuration

The device can be configured with the ReadWin 2000 PC software.

Interface

CDI interface at device; connection to PC via USB box (see "Accessories")

RS232 interface at device; connection with serial interface cable (see "Accessories")

Certificates and approvals

CE mark	The product meets the requirements of the harmonized European standards. As such, it complies with the legal specifications of the EC directives. The manufacturer confirms successful testing of the product by affixing to it the CE-mark.
Ex approval	Information about currently available Ex versions (ATEX, FM, CSA. etc.) can be supplied by your Endress+Hauser sales organization on request. All explosion protection data are given in a separate documentation which is available upon request.
Other standards and guidelines	The manufacturer confirms compliance with all the relevant external standards and guidelines.

Ordering information

Detailed ordering information is available for your nearest sales organization www.addresses.endress.com or in the Product Configurator under www.endress.com :

1. Click Corporate
2. Select the country
3. Click Products
4. Select the product using the filters and search field
5. Open the product page

The Configuration button to the right of the product image opens the Product Configurator.

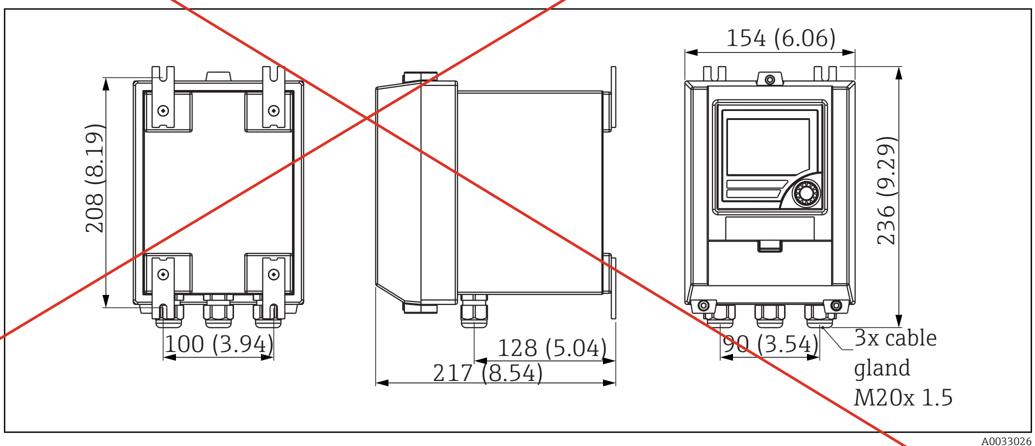
Product Configurator - the tool for individual product configuration
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Accessories

Various accessories, which can be ordered with the device or subsequently from Endress+Hauser, are available for the device. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: www.endress.com.

Device-specific accessories

Designation	Order No.
PC configuration software ReadWin 2000 and serial configuration cable with 3.5 mm jack plug for RS232 port	RIA452A-VK
PC configuration software ReadWin 2000 and serial configuration cable for USB port with CDI connector	TXU10-AA
Field housing in IP65 → 7, 13	51009957
Current simulator active 4-20mA 1-channel, compact housing, 9V block battery	SONDST-S1



7 *Field housing dimensions*

Supplementary documentation

- System components and data manager - solutions to complete your measuring point: FA00016K/09
- Brief Operating Instructions for process indicator RIA452: KA00264R/09
  Operating Instructions for process indicator RIA452: BA00265R/09
- Ex-related additional documentation:
  ATEX II(1)GD: XA00053R/09/a3

www.addresses.endress.com

Endress+Hauser

People for Process Automation